

STAT 200B: Theoretical Statistics

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Statistical decision theory

- A probability model $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Omega\}$ for data $X \in \mathcal{X}$:
 Ω : parameter space, $\mathcal{X}$: sample space.
- An action space $\mathcal{A}$: set of available actions (decisions).
- A loss function:

$$\text{0-1 loss} \quad L(\theta, a) = 1\{\theta \neq a\} \quad \Omega = \mathcal{A} = \{0, 1\}.$$

$$\text{Quadratic loss (Squared error)} \quad L(\theta, a) = \|\theta - a\|_2^2 \quad \Omega = \mathcal{A} = \mathbb{R}^d.$$

Statistical inference as a game:

1. Nature picks the “true” parameter θ , and draws $X \sim \mathbb{P}_\theta$.
Thus, X is a random element of $\mathcal{X}$.
2. Statistician observes X and makes a decision $\delta(X) \in \mathcal{A}$.
 $\delta : \mathcal{X} \rightarrow \mathcal{A}$ is called a *decision rule*.
3. Statistician incurs the loss $L(\theta, \delta(X))$.

The goal of the statistician is to minimize its *expected loss*, a.k.a the *risk*:

$$R(\theta, \delta) := \mathbb{E}_\theta L(\theta, \delta(X))$$

- The goal of the statistician is to minimize its *expected loss*, a.k.a the *risk*:

$$\begin{aligned} R(\theta, \delta) &:= \mathbb{E}_\theta L(\theta, \delta(X)) \\ &= \int L(\theta, \delta(x)) d\mathbb{P}_\theta(x) \\ &= \int L(\theta, \delta(x)) p_\theta(x) d\mu(x) \end{aligned}$$

when family is dominated: $\mathbb{P}_\theta = p_\theta d\mu$.

- Usually work with the family of densities $\{p_\theta(\cdot) : \theta \in \Omega\}$.

Example 1 (Bernoulli trials)

- A coin being flipped, want to estimate the probability of coming up heads.
- One possible model:

$$X = (X_1, \dots, X_n), \quad X_i \stackrel{\text{iid}}{\sim} \text{Ber}(\theta), \quad \text{for some } \theta \in [0, 1].$$

- Formally, $\mathcal{X} = \{0, 1\}^n$, $\mathbb{P}_\theta = (\text{Ber}(\theta))^{\otimes n}$ and $\Omega = [0, 1]$.
- PMF of X_i :

$$\mathbb{P}(X_i = x) = \begin{cases} \theta & x = 1 \\ 1 - \theta & x = 0 \end{cases} = \theta^x (1 - \theta)^{1-x}, \quad x \in \{0, 1\}$$

- Joint PMF: $p_\theta(x_1, \dots, x_n) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}$
- Action space: $\mathcal{A} = \Omega$.
- Quadratic loss: $L(\theta, \delta) = (\theta - \delta)^2$.

Comparing estimators via their risk

Bernoulli trials. Let us look at three estimators:

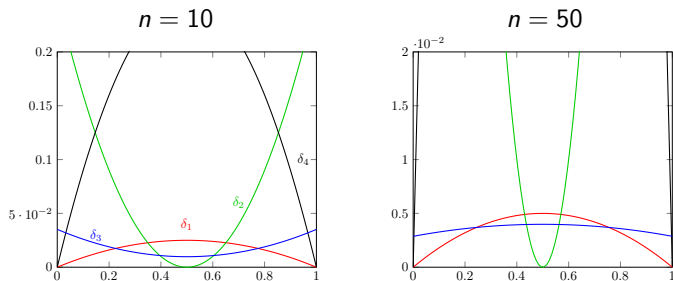
Sample mean	$\delta_1(X) = \frac{1}{n} \sum_{i=1}^n X_i$	$R(\theta, \delta_1) = \frac{\theta(1-\theta)}{n}$
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Constant estimator	$\delta_2(X) = \frac{1}{2}$	$R(\theta, \delta_2) = (\theta - \frac{1}{2})^2$
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Strange looking	$\delta_3(X) = \frac{\sum_i X_i + 3}{n+6}$	$R(\theta, \delta_3) = \frac{n\theta(1-\theta) + (3-6\theta)^2}{(n+6)^2}.$
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Throw data out	$\delta_4(X) = X_1$	$R(\theta, \delta_4) = \theta(1-\theta).$
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Comparing estimators via their risk



Comparison depends on the choice of the loss. A different loss gives a different picture.

Comparing estimators via their risk

How to deal with the fact that the risks are functions?

- Summarize them by reducing to numbers:
 - (Bayesian) Take a weighted average:

$$\inf_{\delta} \int_{\Omega} R(\theta, \delta) d\pi(\theta)$$

- (Frequentist) Take the maximum:

$$\inf_{\delta} \max_{\theta \in \Omega} R(\theta, \delta)$$

- Restrict to a class of estimators: unbiased (UMVU), equivariant, etc.
- Rule out estimators that are dominated by others (inadmissible).

Definition 1

Let δ and δ^* be decision rules. δ^* (strictly) dominates δ if

- $R(\theta, \delta^*) \leq R(\theta, \delta)$, for all $\theta \in \Omega$, and
- $R(\theta, \delta^*) < R(\theta, \delta)$, for some $\theta \in \Omega$.

δ is *inadmissible* if there is a different δ^* that dominates it; otherwise δ is admissible.

An inadmissible rule can be uniformly “improved”.

δ_4 in the Bernoulli example is inadmissible.

We will see a non-trivial example soon (Exponential Distribution).

Definition 2

The bias of δ for estimating $g(\theta)$ is $B_\theta(\delta) := \mathbb{E}_\theta(\delta) - g(\theta)$.

The estimator is *unbiased* if $B_\theta(\delta) = 0$ for all $\theta \in \Omega$.

Not always possible to find unbiased estimators.

Example, $g(\theta) = \sin(\theta)$ in the binomial family. (Keener Example 4.2, p. 62)

Definition 3

g is called *U-estimable* if there an unbiased estimator δ for g .

Usually $g(\theta) = \theta$.

Bias-variance decomposition

For the **quadratic loss** $L(\theta, a) = (\theta - a)^2$, the risk is **mean-squared error (MSE)**. In this case we have the following decomposition

$$\text{MSE}_\theta(\delta) = [B_\theta(\delta)]^2 + \text{var}_\theta(\delta)$$

Proof.

Let $\mu_\theta := \mathbb{E}_\theta(\delta)$. We have

$$\begin{aligned}\text{MSE}_\theta(\delta) &= \mathbb{E}_\theta(\theta - \delta)^2 = \mathbb{E}_\theta(\theta - \mu_\theta + \mu_\theta - \delta)^2 \\ &= (\theta - \mu_\theta)^2 + 2(\theta - \mu_\theta)\mathbb{E}_\theta[\mu_\theta - \delta] + \mathbb{E}_\theta(\mu_\theta - \delta)^2.\end{aligned}$$



Same goes for general $g(\theta)$ or higher dimensions: $L(\theta, a) = \|g(\theta) - a\|_2^2$.

Example 2 (Berger)

Let $X \sim N(\theta, 1)$.

Class of estimators of the form $\delta_c(X) = cX$, for $c \in \mathbb{R}$.

$$\text{MSE}_\theta(\delta) = (\theta - c\theta)^2 + c^2 = (1 - c)^2\theta^2 + c^2$$

For $c > 1$, we have $1 = \text{MSE}_\theta(\delta_1) < \text{MSE}_\theta(\delta_c)$ for all θ .

For $c \in [0, 1]$ the rules are incomparable.

Example 3 (Poisson process)

$X_1, \dots, X_n$ be the inter-arrival times of a Poisson process with rate λ .

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Expo}(\lambda)$. The model has the following p.d.f.

$$p_\lambda(x) = \prod_{i=1}^n p_\lambda(x_i) = \prod_i \lambda e^{-\lambda x_i} 1\{x_i > 0\} = \lambda^n e^{-\lambda \sum_i x_i} 1\{\min_i x_i > 0\}$$

$$\Omega = \mathcal{A} = (0, \infty).$$

- Let $S = \sum_i X_i$ and $\bar{X} = \frac{1}{n}S$.
- The MLE for λ is $\hat{\lambda} = 1/\bar{X} = n/S$.

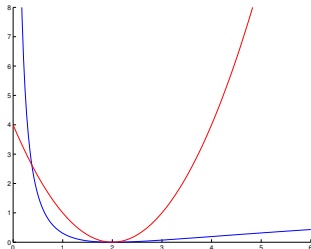
$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Expo}(\lambda)$$

- Let $S = \sum_i X_i$ and $\bar{X} = \frac{1}{n}S$.
- The MLE for λ is $\hat{\lambda} = 1/\bar{X} = n/S$.
- $S := \sum_{i=1}^n X_i$ has the $\text{Gamma}(n, \lambda)$ distribution.
- $1/S$ has $\text{Inv-Gamma}(n, \lambda)$ distribution with mean $\lambda/(n-1)$.
- $\mathbb{E}_\lambda[\hat{\lambda}] = n\lambda/(n-1)$. MLE is biased for λ .
- Then, $\tilde{\lambda} := (n-1)\hat{\lambda}/n$ is unbiased.
- We also have $\text{var}_\lambda(\tilde{\lambda}) < \text{var}_\lambda(\hat{\lambda})$.
- It follows that

$$\text{MSE}_\lambda(\tilde{\lambda}) < \text{MSE}_\lambda(\hat{\lambda}), \quad \forall \lambda$$

The MLE $\hat{\lambda}$ is inadmissible for quadratic loss.

Possible explanation:
 Quadratic loss penalizes over-estimation more than under-estimation for $\Omega = (0, \infty)$.



Alternative loss function (Itakura–Saito distance)

$$L(\lambda, a) = \lambda/a - 1 - \log(\lambda/a), \quad a, \lambda \in (0, \infty)$$

- With this loss function, $R(\lambda, \tilde{\lambda}) > R(\lambda, \hat{\lambda}), \forall \lambda$.
- That is, MLE renders $\tilde{\lambda}$ inadmissible.

An example of a Bregman divergence for $\phi(x) = -\log x$.

For a convex function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, the **Bregman divergence** is defined as

$$d_\phi(x, y) = \phi(x) - \phi(y) - \langle \nabla \phi(y), x - y \rangle$$

the remainder of the first order Taylor expansion of ϕ at y .

Details:

- Consider $\delta_\alpha(X) = \alpha/S$. Then, we have

$$R(\lambda, \delta_\alpha) - R(\lambda, \delta_\beta) = \left(\frac{n}{\alpha} - \frac{n}{\beta} \right) - \left(\log \frac{n}{\alpha} - \log \frac{n}{\beta} \right)$$

- Take $\alpha = n - 1$ and $\beta = n$.
- Use $\log x - \log y < x - y$ for $x > y \geq 1$.

(Follows from strict concavity of $f(x) = \log(x)$:
 $f(x) - f(y) < f'(y)(x - y)$ for $y \neq x$).

Sufficiency

Idea: Separate the data into

- parts that are relevant for the estimating θ (sufficient)
- and parts that are irrelevant (ancillary).

Benefits:

- Achieve data compression: efficient computation and storage
- Irrelevant parts can increase the risk (Rao-Blackwell)

Definition 4

Consider the model $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Omega\}$ for X .

A statistic $T = T(X)$ is sufficient for $\mathcal{P}$ (or for θ or for X) if the conditional distribution of X given T does not depend on θ .

More precisely, we have

$$\mathbb{P}_\theta(X \in A \mid T = t) = \mathbb{Q}_t(A), \quad \forall t, A$$

for some Markov kernel $\mathbb{Q}$. Making it more precise requires measure theory. Intuitively, given T , we can simulate X by an external source of randomness.

Example 4 (Coin tossing)

- $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, $i = 1, \dots, n$.
- Notation: $X = (X_1, \dots, X_n)$, $x = (x_1, \dots, x_n)$.
- Will show that $T = T(X) = \sum_i X_i$ is sufficient for θ . (This should be intuitive.)

$$\mathbb{P}_\theta(X = x) = p_\theta(x) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{T(x)} (1 - \theta)^{n-T(x)}$$

- Then

$$\begin{aligned}\mathbb{P}_\theta(X = x, T = t) &= \mathbb{P}_\theta(X = x) 1\{T(x) = t\} \\ &= \theta^t (1 - \theta)^{n-t} 1\{T(x) = t\}.\end{aligned}$$

- Then

$$\begin{aligned}\mathbb{P}_\theta(X = x, T = t) &= \mathbb{P}_\theta(X = x)1\{T(x) = t\} \\ &= \theta^t(1 - \theta)^{n-t}1\{T(x) = t\}.\end{aligned}$$

- Marginalizing,

$$\begin{aligned}\mathbb{P}_\theta(T = t) &= \theta^t(1 - \theta)^{n-t} \sum_{x \in \{0,1\}^n} 1\{T(x) = t\} \\ &= \binom{n}{t} \theta^t(1 - \theta)^{n-t}.\end{aligned}$$

- Hence,

$$\mathbb{P}_\theta(X = x | T = t) = \frac{\theta^t(1 - \theta)^{n-t}1\{T(x) = t\}}{\binom{n}{t}\theta^t(1 - \theta)^{n-t}} = \frac{1}{\binom{n}{t}}1\{T(x) = t\}.$$

- What is the above (conditional) distribution?

Factorization Theorem

It is not convenient to check for sufficiency this way, hence:

Theorem 1 (Factorization (Fisher–Neyman))

Assume that $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Omega\}$ is dominated by μ . A statistic T is sufficient iff for some function $g_\theta, h \geq 0$

$$p_\theta(x) = g_\theta(T(x))h(x), \quad \text{for } \mu\text{-a.e. } x$$

The likelihood $\theta \mapsto p_\theta(X)$, only depends on X through $T(X)$.

Family being dominated (having a density) is important.

Proof (only discrete case):

Assume $T = T(X)$ is sufficient. Fix x , and let $t = T(x)$, Then,

$$\begin{aligned}\mathbb{P}_\theta(X = x) &= \mathbb{P}_\theta(X = x, T = t) \\ &= \mathbb{P}_\theta(X = x | T = t) \mathbb{P}_\theta(T = t) \\ &= \mathbb{Q}_t(X = x) g_\theta(t).\end{aligned}$$

Factorization Theorem

- Now assume factorization holds. Then,

$$\begin{aligned}\mathbb{P}_\theta(X = x, T = t) &= \mathbb{P}_\theta(X = x)1\{T(x) = t\} \\ &= g_\theta(t)h(x)1\{T(x) = t\}.\end{aligned}$$

- It follows that

$$\mathbb{P}_\theta(T = t) = g_\theta(t) \sum_{x'} h(x')1\{T(x') = t\},$$

- hence

$$\begin{aligned}\mathbb{P}_\theta(X = x|T = t) &= \frac{g_\theta(t)h(x)1\{T(x) = t\}}{g_\theta(t) \sum_{x'} h(x')1\{T(x') = t\}} \\ &= \frac{h(x)1\{T(x) = t\}}{\sum_{x'} h(x')1\{T(x') = t\}}.\end{aligned}$$

Example 5 (Uniform)

- Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta]$.
- Family is dominated by Lebesgue measure.
- $X_{(n)} = \max\{X_1, \dots, X_n\}$ is sufficient by Factorization theorem:

$$\begin{aligned} p_{\theta}(x) &= \prod_{i=1}^n \frac{1}{\theta} 1\{0 \leq x_i \leq \theta\} \\ &= \frac{1}{\theta^n} 1\{0 \leq x_i \leq \theta, \forall i\} = \frac{1}{\theta^n} 1\{0 \leq \min_i x_i\} 1\{\max_i x_i \leq \theta\} \end{aligned}$$

Useful fact $\prod_i 1_{A_i} = 1_{\cap_i A_i}$.

- The entire data $(X_1, \dots, X_n)$ is always sufficient.
- For i.i.d. data there is always more reduction.

Example 6 (IID data)

- Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_\theta$.
- Then, the order statistics $(X_{(1)}, \dots, X_{(n)})$ is sufficient:
- Order the data such that $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$, and discard the **ranks**.

Minimal sufficiency

- There is a hierarchy among sufficient statistics in terms of data reduction.
- Can be made precise by using “functions” as “reduction mechanisms”.

Lemma 1

If T is sufficient for $\mathcal{P}$ and $T = f(S)$ a.e. $\mathcal{P}$, then S is sufficient.

We write $T \leq_s S$ if such functional relation exists.

- An easy consequence of the factorization theorem.

Examples:

- T sufficient $\not\Rightarrow T^2$ sufficient. ($T \not\leq_s T^2$) (Missing sign information)
- T^2 sufficient $\Rightarrow T$ sufficient. ($T^2 \leq_s T$)
- T sufficient $\iff T^3$ sufficient. (bijection)

- $T \leq_s S$ is not standard notation, but useful shorthand for:

$\exists$ function f such that $T = f(S)$ a.e. $\mathcal{P}$.

- We want to obtain a sufficient statistic that achieves greatest reduction, i.e. is at the bottom of that hierarchy.

Definition 5

A statistic T is minimal sufficient if

- T is sufficient, and
- $T \leq_s S$ for any sufficient statistic S .

- Minimal sufficient statistics exist under mild conditions.
- Minimal sufficient statistic is *essentially* unique modulo bijections.

Example 7 (Location family)

Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_\theta$, that is, they have density $p_\theta(x) = f(x - \theta)$. For example consider $f(x) = C \exp(-\beta|x|^\alpha)$.

- The case $\alpha = 2$ corresponds to the normal location family $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Then, $T = \frac{1}{n} \sum_i X_i$ is sufficient by factorization. We will show later that it is minimal sufficient.
- The case $\alpha = 1$ (Laplace or double exponential), no further reduction beyond order statistic is possible.
- A family $\mathcal{P}$ is **DCS** if it dominated with densities having common support.

- Goal: To show that the likelihood (ratio) function is minimal sufficient.
- General idea: For any fixed θ and θ_0 ,

$$\frac{p_{\theta}(X)}{p_{\theta_0}(X)}$$

will always be a function of any sufficient statistic (by Fact. Thm).

- We just have to collect enough of them so that collectively

$$\left(\frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}, \frac{p_{\theta_2}(X)}{p_{\theta_0}(X)}, \frac{p_{\theta_3}(X)}{p_{\theta_0}(X)}, \dots \right)$$

they are sufficient.

A useful lemma

- Let us fix $\theta_0 \in \Omega$ and define

$$L_\theta := L_\theta(X) := \frac{p_\theta(X)}{p_{\theta_0}(X)}.$$

Lemma 2

In a DCS family, the following are equivalent

- (a) U is sufficient for $\mathcal{P}$.
- (b) $L_\theta \leq_s U, \quad \forall \theta \in \Omega$.

- When DCS fails, (a) still implies (b), but not necessarily vice versa.

Proof of Lemma 2

- Work on the common support, densities can be assumed positive.
- (a) $\implies$ (b): U sufficient implies $p_\theta(x) = g_\theta(U(x))h(x)$ (Fact. Thm.):

$$L_\theta(x) = \frac{p_\theta(x)}{p_{\theta_0}(x)} = \frac{g_\theta(U(x))}{g_{\theta_0}(U(x))} = f_{\theta, \theta_0}(U(x))$$

- (b) $\implies$ (a): $\exists f_{\theta, \theta_0}$ such that $p_\theta(x) = p_{\theta_0}(x)f_{\theta, \theta_0}(U(x))$. Q.E.D.

A useful lemma

- Let $\mathbf{L} := (L_\theta)_{\theta \in \Omega}$.
- Since $R \leq_s U$ and $S \leq_s U \iff (R, S) \leq_s U$, we have

Lemma 3

In a DCS family, the following are equivalent

- (a) U is sufficient for $\mathcal{P}$.
- (b) $\mathbf{L} \leq_s U$.

- The argument is correct when Ω is finite.
- Needs more care dealing with “a.e. $\mathcal{P}$ ” when Ω is infinite.
- From Lemma 3 follows that $\mathbf{L}$ is itself sufficient. (Why?)

Proposition 1

In a DCS family, $\mathbf{L} := (L_\theta)_{\theta \in \Omega}$ is minimal sufficient.

Proof:

- Let U be a sufficient statistics.
- Lemma 3(a) $\implies$ (b) gives $\mathbf{L} \leq_s U$.
- (i.e., $\mathbf{L}$ is a function of any sufficient U .)
- We also know that $\mathbf{L}$ is itself sufficient.
- The proof is complete. □

- Consequence of Prop. 1 is

Corollary 1

A statistic T is minimal sufficient if

- *T is sufficient, and*
- *$T \leq_s L$.*

- That is, it is enough to show that T is sufficient and,
- T can be written as a function of L .

$T \leq_s L$ is equivalent to either of the following:

- $L(x) = L(y) \implies T(x) = T(y)$.
- Level sets of L are “included” in level sets of T , i.e.,
- level sets of T are coarser than level set of L .

Corollary 2

T is minimal sufficient for DCS family $\mathcal{P}$ iff

- (a) *T is sufficient, and*
- (b) $\mathbf{L}(x) = \mathbf{L}(y) \implies T(x) = T(y)$

Corollary 3

T is minimal sufficient for DCS family $\mathcal{P}$ iff

$$\mathbf{L}(x) = \mathbf{L}(y) \iff T(x) = T(y)$$

- Can replace $\mathbf{L}(x) = \mathbf{L}(y)$ with $\ell_x(\theta) \propto \ell_y(\theta)$,
where $\ell_x(\theta) = p_\theta(x)$ is the likelihood function. (Theorem 3.11 in Keener.)
- Corollary 3: T is minimal sufficient if it has the same level sets as $\mathbf{L}$.
- Informally, shape of the likelihood is minimal sufficient.

Example 8 (Gaussian location family)

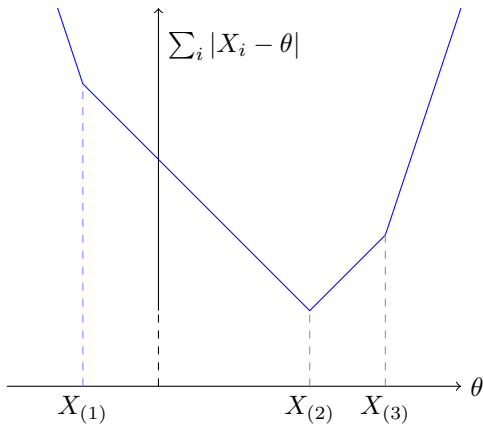
Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_\theta = f(x - \theta)$ with $f(x) = C \exp(-\beta x^2)$.

- $\ell_X(\theta) \propto \exp(-\beta \sum_i (X_i - \theta)^2)$.
- Shape of $\ell_X(\cdot)$ uniquely determined by $\theta \mapsto \sum_i (X_i - \theta)^2$,
- Alternatively, $\theta \mapsto -2(\sum_i X_i)\theta + n\theta^2$,
- Alternatively, $\sum_i X_i$.

Example 9 (Laplace location family)

Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_\theta = f(x - \theta)$ with $f(x) = C \exp(-\beta |x|)$.

- $\ell_X(\theta) \propto \exp(-\beta \sum_i |X_i - \theta|)$.
- Shape of $\ell_X(\cdot)$ uniquely determined by the breakpoints of the piecewise linear function $\theta \mapsto \sum_i |X_i - \theta|$,
- In one-to-one correspondence with the order statistic $(X_{(1)}, \dots, X_{(n)})$.



- Shape of the likelihood for the Laplace location family is determined by the order statistics.

Example with no common support

$\mathcal{P} = \{\mathbb{P}_0, \mathbb{P}_1, \mathbb{P}_2\}$ where

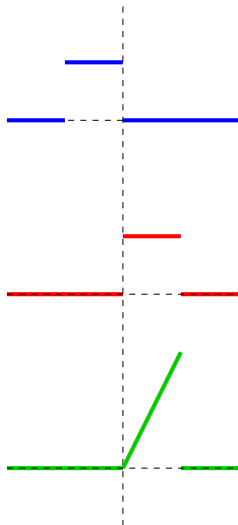
$$\mathbb{P}_0 = U[-1, 0],$$

$$\mathbb{P}_1 = U[0, 1],$$

$$p_2(x) = 2 \times 1\{x \in (0, 1)\}.$$

$$\frac{p_1}{p_0} = \frac{p_2}{p_0} = \begin{cases} 0 & \text{on } (-1, 0) \\ \infty & \text{on } (0, 1) \end{cases}.$$

- Cannot tell $\mathbb{P}_1$ and $\mathbb{P}_2$ based on $(\frac{p_1}{p_0}, \frac{p_2}{p_0})$.
- However, there is information in the original model to statistically tell these two apart to some extent.
- Consider in addition $\frac{p_2(x)}{p_1(x)} = 2 \times 1\{x \in (0, 1)\}$.
- A minimal suff. stat.: $(1\{X < 0\}, X_+)$
- Could just take X_+ , since $1\{X < 0\} = 1 - X_+$

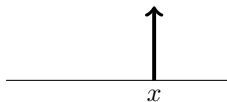


Empirical distribution

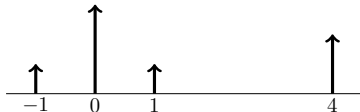
- We saw that for IID data, $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{P}_\theta$,
- the order statistic $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ is sufficient.
- Another way: The **empirical distribution** $\mathbb{P}_n$ is **sufficient**,

$$\mathbb{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}, \quad (\delta_x: \text{unit point mass at } x)$$

- δ_x is a measure defined by: $\delta_x(A) := 1\{x \in A\}$:



- Example: $X = (0, 1, -1, 0, 4, 4, 0) \implies \mathbb{P}_n := \frac{1}{7}(\delta_{-1} + 3\delta_0 + \delta_1 + 2\delta_4)$.



Example 10 (Empirical distribution in finite alphabet (IID data))

- Suppose sample space is finite $\mathcal{X} = \{a_1, \dots, a_K\}$.
- Let $\mathcal{P} = \{ \text{collection of all prob. measures } \mathbb{P} \text{ on } \mathcal{X} \}$.
- $\mathcal{P}$ can be parametrized by $\theta = (\theta_1, \dots, \theta_K)$ where $\theta_j = \mathbb{P}(\{a_j\})$.
- Empirical measure reduces to $\mathbb{P}_n = \sum_j \pi_j(X) \delta_{a_j}$ where

$$\pi_j(X) := \frac{1}{n} \# \{i : X_i = a_j\}$$

- Show that $\mathbb{P}_n$ or equivalently $(\pi_1(X), \dots, \pi_K(X))$ is minimal sufficient.

Statements from Theory of Point Estimation (TPE)

Proposition 2 (TPE)

Consider a finite DCS family $\mathcal{P} = \{\mathbb{P}_\theta, \theta \in \Omega\}$, i.e., $\Omega = \{\theta_0, \theta_1, \dots, \theta_K\}$. Then, the following is minimal sufficient

$$\mathbf{L}(X) = \left(\frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}, \dots, \frac{p_{\theta_K}(X)}{p_{\theta_0}(X)} \right).$$

Proposition 3 (TPE)

Assume $\mathcal{P}$ is DCS and $\mathcal{P}_0 \subset \mathcal{P}$. Assume that T is

- sufficient for $\mathcal{P}$, and
- minimal sufficient for $\mathcal{P}_0$.

Then, T is minimal sufficient for $\mathcal{P}$.

Same support gives “a.e. $\mathcal{P}_0 \implies$ a.e. $\mathcal{P}$ ”.

S sufficient for $\mathcal{P} \implies S$ sufficient for $\mathcal{P}_0$.

T minimal suff. for $\mathcal{P}_0 \implies T = f(S)$ a.e. $\mathcal{P}_0$ and hence a.e. $\mathcal{P}$. Q.E.D.

Example 11 (Gaussian location family)

- Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$.
- Look at sub-family $\mathcal{P}_0 = \{N(\theta_0, 1), N(\theta_1, 1)\}$. Let $T(X) = \sum_i X_i$.
- The following is minimal sufficient by Proposition 2,

$$\begin{aligned}\log L_{\theta_1} &:= \log \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)} = \frac{1}{2} \sum_i (x_i - \theta_0)^2 - \frac{1}{2} \sum_i (x_i - \theta_1)^2 \\ &= (\theta_1 - \theta_0) T(x) + \frac{1}{2} (\theta_0^2 - \theta_1^2)\end{aligned}$$

showing that $T(X)$ is minimal sufficient for $\mathcal{P}_0$.

- Since $T(X)$ is also sufficient for $\mathcal{P}$ (Exercise.), Proposition 3 implies that it is minimal sufficient for $\mathcal{P}$.

Completeness and ancillarity

- We can compress even more!

Definition 6

- $V = V(X)$ is ancillary if its distribution does not depend on θ .
- First-order ancillary if its expectation does not depend on θ . ($\mathbb{E}_\theta V = c, \forall \theta$).
- The latter is a weaker notion.

Example 12 (Location family)

- The following statistics are all ancillary:

$$X_i - X_j, \quad X_i - X_{(j)}, \quad X_{(i)} - X_{(j)}, \quad X_{(i)} - \bar{X}$$

- Hint: We can write $X_i = \theta + \varepsilon_i$, where $\varepsilon_i \stackrel{\text{iid}}{\sim} f$.
- Minimal sufficient statistic can still contain ancillary information, for example $X_{(n)} - X_{(1)}$ in the Laplace location family.

- A notion stronger than minimal sufficiency is completeness:

Definition 7

A statistic T is complete if

$$(\mathbb{E}_\theta[f(T)] = c, \forall \theta) \implies f(t) = c \quad \forall t.$$

(Actually $\mathcal{P}$ -a.e. t .)

- T is complete if no nonconstant function of it is first-order ancillary.
- Minimal sufficient statistic need not be complete:

Example 13 (Laplace location family)

- $X_{(n)} - X_{(1)}$ is ancillary, hence first-order ancillary.
- $f(X_{(1)}, \dots, X_{(n)})$ is ancillary for the nonconstant function $f(z_1, \dots, z_n) = z_1 - z_n$.
- The converse is however true.

- Showing completeness is not easy.
- Will see a general result for exponential families.
- Here is another example:

Example 14

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta]$.
- Will show that $T = \max\{X_1, \dots, X_n\}$ is complete.
- CDF of T is $F_T(t) = (t/\theta)^n 1\{t \in (0, \theta)\} + 1\{t \geq \theta\}$.
- Density $t \mapsto n\theta^{-n}t^{n-1}1\{t \in (0, \theta)\}$.
- Suppose that $\mathbb{E}_\theta f(T) = 0$ for all $\theta > 0$. Then,

$$0 = \mathbb{E}_\theta f(T) = n\theta^{-n} \int_0^\theta f(t)t^{n-1}dt, \quad t > 0$$

- Fundamental theorem of calculus implies $f(t)t^{n-1} = 0$, a.e. $t > 0$,
- Hence $f(t) = 0$ a.e. $t > 0$. Conclude that T is complete.

Detour: Conditional expectation as L^2 projection

- The L^2 space of random variables:

$$L^2 := L^2(\mathbb{P}) := \{X : \mathbb{E}[X^2] < \infty\}$$

- We can define an inner product on this space

$$\langle X, Y \rangle := \mathbb{E}[XY], \quad X, Y \in L^2$$

- The inner product induces a norm, called the L^2 norm,

$$\|X\|_2 := \sqrt{\langle X, X \rangle} := \sqrt{\mathbb{E}[X^2]}$$

- Norm induces a distance $\|X - Y\|_2$.
- Squared distance $\|X - Y\|_2^2 = \mathbb{E}(X - Y)^2$, the same as MSE.
- Orthogonality: $X \perp Y$ if $\langle X, Y \rangle = 0$, i.e., $\mathbb{E}[XY] = 0$.

Detour: Conditional expectation as L^2 projection

- Assume $\mathbb{E}X^2 < \infty$ and $\mathbb{E}Y^2 < \infty$ (i.e., $X, Y \in L^2$).
- Consider the linear space

$$\mathcal{L} := \{g(X) \mid g \text{ is a (measurable) function with } \mathbb{E}[g(X)]^2 < \infty\}$$

i.e., the space of all (meas.) functions of X .

- There is an essentially unique L^2 projection of Y onto $\mathcal{L}$:

$$\hat{Y} := \operatorname{argmin}_{Z \in \mathcal{L}} \|Y - Z\|_2$$

- Alternatively, an essentially unique function $\hat{g}$ such that

$$\min_g \mathbb{E}(Y - g(X))^2 = \mathbb{E}(Y - \hat{g}(X))^2$$

- We define $\mathbb{E}[Y|X] := \hat{g}(X)$.

Detour: Conditional expectation as L^2 projection

- There is an essentially unique function $\hat{g}$ such that

$$\min_g \mathbb{E}(Y - g(X))^2 = \mathbb{E}(Y - \hat{g}(X))^2$$

- We define $\mathbb{E}[Y|X] := \hat{g}(X)$.
- $\mathbb{E}[Y|X]$ is the best prediction of Y given X , in the MSE sense.
- From this definition, we get the following characterization of $\hat{g}$:

$$\mathbb{E}[(Y - \hat{g}(X))g(X)] = 0, \quad \forall g$$

saying that the optimal prediction error $Y - \hat{g}(X)$ is orthogonal to $\mathcal{L}$.

- Applied to the constant function $g(X) \equiv 1$, we get

$$\mathbb{E}[Y] = \mathbb{E}[\hat{g}(X)] = \mathbb{E}[\mathbb{E}[Y|X]]$$

the smoothing or averaging property of conditional expectation.

Proposition 4

A complete sufficient statistic is minimal sufficient.

Proof. Let T be complete sufficient, and U minimal sufficient.

- Idea is to show that T is a function of U .
- $U = g(T)$. (By minimal sufficiency of U .)
- Let $h(U) := \mathbb{E}_\theta[T|U]$ well defined by sufficiency of U .
- $\mathbb{E}_\theta[T - h(U)] = 0, \forall \theta \in \Omega$. (By smoothing.)
- $T = h(U)$. (By completeness of T .)

Hints:

- Took $f(t) := t - h(g(t))$ in the definition of completeness.
- Equalities are a.e. $\mathcal{P}$.

Proposition 5 (Basu)

*Let T be complete sufficient and V ancillary.
Then T and V are independent.*

Proof. Let A be an event.

- $q_A := \mathbb{P}_\theta(V \in A)$ well-defined. (By ancillary of V .)
- $f_A(T) := \mathbb{P}_\theta(V \in A | T)$ well-defined. (By sufficiency of T .)
- $\mathbb{E}_\theta[q_A - f_A(T)] = 0, \forall \theta.$
- $q_A = f_A(T).$ (By completeness of T .)

Equalities are a.e. $\mathcal{P}$.

- Application of Basu:

Example 15 (Gaussian location family)

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$, θ is unknown, σ^2 is known.
 - $\bar{X} := \frac{1}{n} \sum_i X_i$ is complete sufficient. (cf. Exponential families)
 - $(X_i - \bar{X}, i = 1, \dots, n)$ is ancillary.
 - Hence, sample variance $S^2 := \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$ is ancillary.
 - Hence, $\bar{X}$ and S^2 are independent.
-
- Had we taken (θ, σ^2) as the parameter, then S^2 would not be ancillary.

- Rao–Blackwell theorem ties risk minimization with sufficiency.
- It is a statement about convex loss functions.

Definition 8

A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is convex if for all x, y

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \forall \lambda \in [0, 1], \quad (1)$$

and strictly convex if the inequality is strict for $\lambda \in (0, 1)$ and $x \neq y$.

Example 16 (ℓ_p loss)

- Loss function $a \mapsto L(\theta, a) = |\theta - a|^p$ on $\mathbb{R}$.
- Convex for $p \geq 1$ and nonconvex for $p \in (0, 1)$.
- Strictly convex when $p > 1$.
- In particular, the ℓ_1 loss ($p = 1$) is convex but not strictly convex.

Jensen inequality

By induction (1) leads to $f(\sum_i \alpha_i x_i) \leq \sum_i \alpha_i f(x_i)$, for $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$.
A sweeping generalization is the following:

Proposition 6 (Jensen inequality)

Assume that $f : S \rightarrow \mathbb{R}$ is convex and consider a random variable X concentrated on S (i.e., $\mathbb{P}(X \in S) = 1$), and $\mathbb{E}|X| < \infty$. Then,

$$\mathbb{E}f(X) \geq f(\mathbb{E}X)$$

If f is strictly convex, equality holds iff $X \equiv \mathbb{E}X$ a.s. (that is, X is constant).

Proof. Relies on the existence of supporting hyperplanes to f (i.e., affine minorants that touch the function)

- Let $x_0 := \mathbb{E}X$.
- Let $A(x) = \langle a, x - x_0 \rangle + f(x_0)$ be a supporting hyperplane to f at x_0 :

$$f(x) \geq A(x), \quad \forall x \in \mathcal{S}, \quad \text{and} \quad A(x_0) = f(x_0).$$

- Then, we have

$$\begin{aligned} f(X) \geq A(X) &\implies \mathbb{E}[f(X)] \geq \mathbb{E}[A(X)] && \text{(Monotonicity of } \mathbb{E}) \\ &= \langle a, \mathbb{E}[X - x_0] \rangle + f(x_0) && \text{(Linearity of } \mathbb{E}) \\ &= f(x_0) \end{aligned}$$

- Strict convexity implies $f(x) > A(x)$ for $x \neq x_0$.
- If $X \neq x_0$ with positive prob., we have $f(X) > A(X)$ with positive prob.,
- Hence, $\mathbb{E}f(X) > \mathbb{E}A(X)$, and the rest follows. Q.E.D.

Recall the decision-theoretic setup:

Loss function $L(\theta, a)$, decision rule $\delta = \delta(X)$, risk $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta)$.

Theorem 2 (Rao–Blackwell)

Let us assume the following:

- T is sufficient for family $\mathcal{P}$,
- $\delta = \delta(X)$ is a possibly randomized decision rule,
- $a \mapsto L(\theta, a)$ is convex, for all $\theta \in \Omega$.

Define the estimator $\eta(T) := \mathbb{E}_\theta[\delta|T]$. Then, η dominates δ , that is,

$$R(\theta, \eta) \leq R(\theta, \delta) \text{ for all } \theta \in \Omega$$

The inequality is strict when the loss is strictly convex, unless $\eta = \delta$ a.e. $\mathcal{P}$.

Consequence: for convex loss functions, randomization does not help. Proof:

- η is well-defined. (By sufficiency of T .)
- $\mathbb{E}_\theta[L(\theta, \delta)|T] \geq L(\theta, \mathbb{E}_\theta[\delta|T]) = L(\theta, \eta)$. (By conditional Jensen inequality.)
- Take expectation and use monotonicity and smoothing. Q.E.D.

Example 17

- $X_i \stackrel{\text{iid}}{\sim} U[0, \theta], i = 1, \dots, n.$
- $T = \max\{X_1, \dots, X_n\}$ is sufficient. Take $\delta = \frac{1}{n} \sum_i X_i.$
- Rao–Blackwell: $\eta = \mathbb{E}_\theta[\delta | T]$ strictly dominates δ , for any strictly convex loss. Let us verify this:
- Conditional distribution of X_i given T :
Mixture of a point mass at T and uniform distribution on $[0, T]$ (why?):

$$\mathbb{P}_\theta(X_i \in A | T) = \frac{1}{n} \delta_T(A) + \left(1 - \frac{1}{n}\right) \int_0^T \frac{1}{T} 1_A(x) dx$$

- Compactly $X_i | T \sim \frac{1}{n} \delta_T + (1 - \frac{1}{n}) \text{Unif}(0, T).$
- It follows

$$\mathbb{E}_\theta(X_i | T) = \frac{1}{n} T + \left(1 - \frac{1}{n}\right) \frac{T}{2} = \frac{n+1}{2n} T$$

- Same expression holds for η by symmetry.

- That is, $\eta = \mathbb{E}_\theta[\delta | T] = \frac{n+1}{2n} T$
- Consider the quadratic loss:
- η has the same bias as δ (by smoothing).
- How about variances?

$$\text{var}_\theta(\delta) = \frac{1}{n} \frac{\theta^2}{12}$$

- Since T/θ is Beta($n, 1$) distributed, we have

$$\text{var}_\theta(\eta) = \left(\frac{n+1}{2n}\right)^2 \frac{n}{(n+1)^2(n+2)} \theta^2 = \frac{\theta^2}{4n(n+2)} \quad (2)$$

- (Note: δ is biased. 2δ is unbiased. Better to compare 2η and 2δ .)

- Another result about strictly convex loss functions:
An admissible decision rule is uniquely determined by its risk function.

Proposition 7

• Assume $a \mapsto L(\theta, a)$ is strictly convex, for all θ . ($R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta)]$.)
Then, the map $\delta \mapsto R(\cdot, \delta)$ is injective over the class of admissible decision rules.

We are identifying decision rules that are the same a.e. $\mathcal{P}$.

Proof.

- Let δ be admissible.
- Let $\delta' \neq \delta$ be such that $R(\theta, \delta) = R(\theta, \delta'), \forall \theta$.
- Take $\delta^* = \frac{1}{2}(\delta + \delta')$. Then, by strict convexity of the loss

$$L(\theta, \delta^*) < \frac{1}{2}(L(\theta, \delta) + L(\theta, \delta')), \forall \theta$$

Taking expectation: (Note: $X > 0$ a.s. $\implies \mathbb{E}[X] > 0$)

$$R(\theta, \delta^*) < \frac{1}{2}(R(\theta, \delta) + R(\theta, \delta')) = R(\theta, \delta), \forall \theta$$

- $\delta' \neq \delta$ implies $\delta^* \neq \delta$.
- δ^* strictly dominates δ , contradicting admissibility of δ .

Rao–Blackwell can fail for non-convex loss. Here is an example:

Example 18

- $X \sim \text{Bin}(n, \theta)$. $\Omega = \mathcal{A} = [0, 1]$.
 - ε -sensitive loss function $L_\varepsilon(\theta, a) = 1\{|\theta - a| \geq \varepsilon\}$.
 - Consider a general deterministic estimator of $\delta = \delta(X)$.
 - δ takes at most $n + 1$ values $\{\delta(0), \delta(1), \dots, \delta(n)\} \subset [0, 1]$.
 - Divide $[0, 1]$ into bins of length 2ε .
 - Assume that $N := 1/(2\varepsilon) \geq n + 2$ and that N is an integer (for simplicity).
- At least one of the N bins contains no $\delta(i), i = 0, \dots, n$, and the midpoint of that bin is at distance $\geq \varepsilon$ from any $\delta(i)$. Hence

$$\sup_{\theta \in [0, 1]} R(\theta, \delta) = 1$$

for any nonrandomized rule (assuming $\varepsilon \leq 1/[2(n + 2)]$).

- Consider randomized estimator $\delta' = U \sim U[0, 1]$ independent of X :

$$R(\theta, \delta') = \mathbb{P}(|U - \theta| \geq \varepsilon) \leq 1 - \varepsilon \quad (\text{worst case at } \theta = 0, 1.)$$

- $\sup_{\theta \in [0, 1]} R(\theta, \delta') < 1$ strictly better than that of δ .

Uniformly minimum variance unbiased (UMVU) criterion

- Comparing estimators based on their risk functions problematic.
- One way to mitigate: restrict the class of estimators.
- Focus on quadratic loss, restrict to **unbiased** estimators.
- Bias-variance decomp.,

Let $\mathcal{U}_g$ be the class of unbiased estimators of $g(\theta)$, that is,

$$\mathcal{U}_g = \{\delta : \mathbb{E}_\theta[\delta] = g(\theta), \forall \theta\}.$$

Definition 9

An estimator δ is UMVU for estimating $g(\theta)$ if

- $\delta \in \mathcal{U}_g$, and
- $\text{var}_\theta(\delta) \leq \text{var}_\theta(\delta')$, for $\theta \in \Omega$, and for all $\delta' \in \mathcal{U}_g$.

Theorem 3 (Lehmann–Scheffé)

Consider the family $\mathcal{P}$ and assume that

- $\mathcal{U}_g$ is nonempty (i.e., g is U -estimable), and
- there is a complete sufficient statistic T for $\mathcal{P}$.

Then, there is an essentially unique UMVU for $g(\theta)$ of the form $h(T)$.

Proof.

- Pick $\delta \in \mathcal{U}_g$ (Valid by non-emptiness.)
- Let $\eta = \mathbb{E}_\theta[\delta | T]$ be an estimator (Well-defined by sufficiency of T .)
- Claim: η is the essentially unique UMVU.
- Pick any $\delta' \in \mathcal{U}_g$ and let $\eta' = \mathbb{E}_\theta[\delta' | T]$.
- $\mathbb{E}_\theta[\eta - \eta'] = g(\theta) - g(\theta) = 0, \forall \theta$ (By smoothing and unbiasedness.)
- (a) $\eta - \eta' = 0$ a.e. $\mathcal{P}$. (By completeness of T .)
- By Rao–Blackwell for quadratic loss $a \mapsto (g(\theta) - a)^2$ and unbiasedness

$$\text{var}_\theta(\eta) = R(\theta, \eta) \stackrel{(a)}{=} R(\theta, \eta') \leq R(\theta, \delta') = \text{var}_\theta(\delta')$$

- Since δ' was an arbitrary element of the class $\mathcal{U}_g$ we are done. Q.E.D.

Proof.

- Pick $\delta \in \mathcal{U}_g$ (Valid by non-emptiness.)
- Let $\eta = \mathbb{E}_\theta[\delta | T]$ be an estimator (Well-defined by sufficiency of T .)
- Claim: η is the essentially unique UMVU.
- Pick any $\delta' \in \mathcal{U}_g$ and let $\eta' = \mathbb{E}_\theta[\delta' | T]$.
- $\mathbb{E}_\theta[\eta - \eta'] = g(\theta) - g(\theta) = 0, \forall \theta$ (By smoothing and unbiasedness.)
- $\eta - \eta' = 0$ a.e. $\mathcal{P}$. (By completeness of T .)
- By Rao–Blackwell for quadratic loss $a \mapsto (g(\theta) - a)^2$ and unbiasedness

$$\text{var}_\theta(\eta) = R(\theta, \eta) = R(\theta, \eta') \stackrel{(b)}{<} R(\theta, \delta') = \text{var}_\theta(\delta')$$

- Since δ' was an arbitrary element of the class $\mathcal{U}_g$ we are done. Q.E.D.

Remark 1

Note that we have also shown the uniqueness:

- (b) If $\delta' \in \mathcal{U}_g$ is UMVU and not a function of T , then it is strictly dominated by η' (by Rao–Blackwell and strict convexity of quadratic loss).
- Otherwise, it is equal to η' which is equal to η a.e. $\mathcal{P}$.

Lehman–Schuffé suggest a way of constructing UMVUs.

Example 19 (Coin tossing)

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, want to estimate $g(\theta) = \theta^2$.
- $T = \sum_i X_i$ is complete and sufficient.
(General result for exponential families.)
- Take $U = X_1 X_2$.
- U is unbiased for θ^2 : $\mathbb{E}_\theta[U] = \mathbb{E}_\theta[X_1] \mathbb{E}_\theta[X_2] = \theta^2$ by independence.
- By Lehman–Schuffé

$$\begin{aligned} \mathbb{E}[U | T = t] &= \mathbb{P}(X_1 + X_2 = 2 | T = t) \\ &= \begin{cases} \binom{n-2}{t-2} / \binom{n}{t}, & t \geq 2 \\ 0 & \text{otherwise} \end{cases} = \frac{t(t-1)}{n(n-1)} \end{aligned}$$

is UMVU estimator for θ^2 .

Approach 2 in obtaining UMVUs:

Example 20

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta]$.
- $T = X_{(n)} = \max\{X_1, \dots, X_n\}$ is complete sufficient.
- UMVU for $g(\theta)$ is given by $h(X_{(n)})$.
- h is the solution of the following integral equation:

$$g(\theta) = \mathbb{E}_\theta[h(X_{(n)})] = n\theta^{-n} \int_0^\theta t^{n-1} h(t) dt.$$

- For $g(\theta) = \theta$, $\delta_1 = \frac{n+1}{n} T$ is unbiased, hence UMVU by Lehmann–Schuffé.
- $\text{MSE}_\theta(\delta_1) = \text{var}_\theta(\delta_1) = \frac{\theta^2}{n(n+2)}$.
- On the other hand, among estimators of the form $\delta_a = aT$, $a = (n+2)/(n+1)$ gives the lowest MSE.
- This biased estimator has slightly better $\text{MSE} = \theta^2/(n+1)^2$.
- A little bit of bias is not bad.

Exponential family

Definition 10

$\mathcal{X}$: a general sample space, Ω : a general parameter space,

- A function $T : \mathcal{X} \rightarrow \mathbb{R}^d$, $T(x) = (T_1(x), \dots, T_d(x))$.
- A function $\eta : \Omega \rightarrow \mathbb{R}^d$, $\eta(\theta) = (\eta_1(\theta), \dots, \eta_d(\theta))$.
- A measure ν on $\mathcal{X}$ (e.g., Lebesgue or counting), and a function $h : \mathcal{X} \rightarrow \mathbb{R}_+$.

The exponential family with **sufficient statistic** T and parametrization η , relative to $h \cdot \nu$, is the dominated family of distributions given by the following densities w.r.t. ν

$$p_\theta(x) = \exp \{ \langle \eta(\theta), T(x) \rangle - A(\theta) \} h(x), \quad x \in \mathcal{X}.$$

where $\langle \eta(\theta), T(x) \rangle = \sum_{i=1}^d \eta_i(\theta) T_i(x)$ is the Euclidean inner product.

- $T : \mathcal{X} \rightarrow \mathbb{R}^d, \quad \eta : \Omega \rightarrow \mathbb{R}^d.$

$$p_{\theta}(x) = \exp \left\{ \langle \eta(\theta), T(x) \rangle - A(\theta) \right\} h(x), \quad x \in \mathcal{X}.$$

- $A(\theta)$ is determined by the other ingredients,
- via the normalization constraint $\int p_{\theta}(x) d\nu(x) = 1,$

$$A(\theta) = \log \int e^{\langle \eta(\theta), T(x) \rangle} d\tilde{\nu}(x)$$

where $d\tilde{\nu}(x) := h(x)d\nu(x).$

- A is called the **log partition function** or **cumulant generating function**.
- The actual parameter space is

$$\Omega_0 = \{\theta \in \Omega : A(\theta) < \infty\}.$$

- By factorization theorem, $T(X)$ is indeed sufficient.
- The representation of the exponential family is not unique.

Here are some examples:

Example 21

- $X \sim \text{Ber}(\theta)$:

$$p_{\theta}(x) = \theta^x(1 - \theta)^{1-x} = \exp \left[x \log \frac{\theta}{1 - \theta} + \log(1 - \theta) \right].$$

- Here $h(x) = 1\{x \in \{0, 1\}\}$,

$$\eta(\theta) = \log \left(\frac{\theta}{1 - \theta} \right), \quad T(x) = x$$

$$A(\theta) = -\log(1 - \theta), \quad \Omega_0 = (0, 1)$$

- We need to take $\Omega = (0, 1)$ otherwise η is not well-defined.

Example 22

- $X \sim N(\mu, \sigma^2)$: Let $\theta = (\mu, \sigma^2)$,

$$\begin{aligned} p_{\theta}(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2} (x - \mu)^2 \right] \\ &= \exp \left[-\frac{1}{2\sigma^2} x^2 + \frac{\mu}{\sigma^2} x - \left(\frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2) \right) \right] \end{aligned}$$

- Here, $h(x) = 1$,

$$\begin{aligned} \eta(\theta) &= \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right), & T(x) &= (x, x^2) \\ A(\theta) &= \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2), & \Omega_0 &= \{(\mu, \sigma^2) : \sigma^2 > 0\} \end{aligned}$$

- We could have taken $h(x) = \frac{1}{\sqrt{2\pi}}$ and $A(\theta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log \sigma^2$.

Example 23

- $X \sim U[0, \theta]$,
- $p_{\theta}(x) = \theta^{-1} \mathbf{1}\{x \in (0, \theta)\}$.
- Not an exponential family since the support depends on the parameter.

Consider the following conditions:

(E1) $\eta(\Omega_0)$ has non-empty interior.

(E2) $\{T_1, \dots, T_d, 1\}$ are linearly independent $\tilde{\nu}$ a.e.. That is,

$$\nexists a \in \mathbb{R}^d \setminus \{0\}, c \in \mathbb{R} \quad \text{such that} \quad \langle a, T(x) \rangle = c, \quad \tilde{\nu}\text{-a.e. } x$$

(E1') $\eta(\Omega_0)$ is open.

Definition 11

- A family satisfying (E1) and (E2) is called full-rank.
 - One that satisfies (E1') is regular.
 - One that satisfies (E2) is minimal.
-
- Condition (E1) prevents $\{\eta_i\}$ from satisfying a constraint.
 - Condition (E2) prevents **unidentifiability**.

Example 24

- A Bernoulli model: $p_{\theta}(x) \propto \exp(\theta_0(1 - x) + \theta_1 x)$.
- $x + (1 - x) = 1, \forall x$. Hence, the family is not full-rank.

Example 25

- A continuous model with $\Omega = \mathbb{R}$ and $\eta_1(\theta) = \theta$, $\eta_2(\theta) = \theta^2$,
- Interior of $\eta(\Omega)$ is empty, hence the model is not full-rank.

Theorem 4

In a full-rank exponential family, T is complete.

- We just show that T is minimal sufficient.
- Completeness is more technical, but follows from Laplace transform arguments.

Theorem 5

In a full-rank exponential family, T is complete.

Proof. (Minimal sufficiency.)

- By factorization theorem, T is sufficient for $\mathcal{P}$ (the whole family).
- Choose $\{\theta_0, \theta_1, \dots, \theta_d\} \subset \Omega$, with $\eta_i := \eta(\theta_i)$, such that

$\{\eta_1 - \eta_0, \eta_2 - \eta_0, \dots, \eta_d - \eta_0\}$ are linearly independent. Possible by (E1)

- Matrix $A^T = (\eta_1 - \eta_0, \dots, \eta_d - \eta_0) \in \mathbb{R}^{d \times d}$ is full-rank.
- Let $\mathcal{P}_0 = \{p_{\theta_i} : i = 0, 1, \dots, d\}$. Then, with $T = T(X)$

$$\left(\log \frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}, \dots, \log \frac{p_{\theta_d}(X)}{p_{\theta_0}(X)} \right) = (\langle \eta_1 - \eta_0, T \rangle, \dots, \langle \eta_d - \eta_0, T \rangle) = AT$$

is minimal sufficient for $\mathcal{P}_0$.

- It follows that T is so, since A is invertible.
- Since $\mathcal{P}_0$ and $\mathcal{P}$ have common support, T is also minimal for $\mathcal{P}$.

$$p_{\theta}(x) = \exp \{ \langle \eta(\theta), T(x) \rangle - A(\theta) \} h(x), \quad x \in \mathcal{X}.$$

Definition 12

An exponential family is in canonical (or natural) form if $\eta(\theta) = \theta$.

In this case:

- $\eta = \theta$ is called the natural parameter.
- $\Omega_0 := \{ \theta \in \mathbb{R}^d : A(\theta) < \infty \}$ is called the natural parameter space.
- $\Omega_0 \subset \mathbb{R}^d$.

Family determined by choice of $\mathcal{X}$, $T(x)$ and $\tilde{\nu} = h \cdot \nu$.

Example 26 (Two-parameter Gaussian)

- $\mathcal{X} = \mathbb{R}$, $T(x) = (x, x^2)$.
- $p_{\theta}(x) = \exp(\theta_1 x + \theta_2 x^2 - A(\theta))$, $\forall x \in \mathcal{X}$.
- $A(\theta) = \log \int e^{\theta_1 x + \theta_2 x^2} dx$.
- $A(\theta) < \infty$ iff $\theta_2 < 0$. Natural parameter space: $\Omega_0 = \{(\theta_1, \theta_2) : \theta_2 < 0\}$.
- Note: $\theta_1 = \frac{\mu}{\sigma^2}$ and $\theta_2 = -\frac{1}{2\sigma^2}$ (in original parametrization (μ, σ^2)).

Recall: $\|x\|_1 = \sum_{i=1}^d |x_i|$.

Example 27 (Multinomial)

- $\mathcal{X} = \mathbb{Z}_+^d = \{x = (x_1, \dots, x_d) : x_i \text{ integer, } x_i \geq 0\}$
- $T(x) = x$.
- $h(x) = \binom{n}{x_1, x_2, \dots, x_d} 1\{\|x\|_1 = n\}$, $\nu = \text{counting measure}$
- Canonical family: $p_\theta(x) = \exp\left(\sum_{i=1}^d \theta_i x_i - A(\theta)\right) h(x)$.
- Can show that $A(\theta) = n \log\left(\sum_{i=1}^d e^{\theta_i}\right)$, finite everywhere.
- Hence $\Omega_0 = \mathbb{R}^d$.
- Not full-rank. Violates (E2).

- Multinomial distribution, with usual parameter,

$$q_{\pi}(x) = \binom{n}{x_1, x_2, \dots, x_d} \prod_{i=1}^d \pi_i^{x_i}$$

looks like a subfamily:

- Corresponds to the following subset of the natural parameter space Ω_0 :

$$\left\{ (\log \pi_i) : \pi_i > 0, \sum_{i=1}^d \pi_i = 1 \right\} = \left\{ \theta \in \mathbb{R}^d : \sum_{i=1}^d e^{\theta_i} = 1 \right\}.$$

- This family is also not full-rank (violates (E1)).
- Actually not a sub-family of Example 27 since $p_{\theta} = p_{\theta+a\mathbf{1}}$ for any $a \in \mathbb{R}$.
- That is, θ parametrization is **non-identifiable** in Example 27.

Example 28 (Multivariate Gaussian)

- $\mathcal{X} = \mathbb{R}^p$. $\tilde{\nu}$ = Lebesgue measure on $\mathcal{X}$.
- $T(x) = (x_i, i = 1, \dots, p \mid x_i x_j, 1 \leq i < j \leq p \mid x_i^2, i = 1, \dots, p)$.
- $\text{param} = (\theta_i, i = 1, \dots, p \mid 2\Theta_{ij}, 1 \leq i < j \leq p \mid \Theta_{ii}, i = 1, \dots, p)$.
- Corresponding canonical Expf

$$p_{\theta, \Theta}(x) = \exp \left(\sum_i \theta_i x_i + 2 \sum_{i < j} \Theta_{ij} x_i x_j + \sum_i \Theta_{ii} x_i^2 - A(\theta, \Theta) \right)$$

- Compactly, treating Θ as a symmetric matrix,

$$p_{\theta, \Theta}(x) = \exp \{ \langle \theta, x \rangle + \langle \Theta, xx^T \rangle - A(\theta, \Theta) \}$$

where $\langle \Theta, xx^T \rangle := \text{tr}(\Theta xx^T) = \text{tr}(x^T \Theta x) = x^T \Theta x$.

- Dimension (or rank) of the family $d = p + p(p-1)/2$.

- Density of multivariate Gaussian $N(\mu, \Sigma)$:

$$\begin{aligned} p_{\mu, \Sigma}(x) &\propto \frac{1}{|\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right] \\ &= \exp \left[-\frac{1}{2} x^T \Sigma^{-1} x + x^T \Sigma^{-1} \mu - \frac{1}{2} \mu^T \Sigma^{-1} \mu - \frac{1}{2} \log |\Sigma| \right] \end{aligned}$$

- Can be written as a canonical exponential family

$$p_{\theta, \Theta}(x) = \exp \{ \langle \theta, x \rangle + \langle \Theta, xx^T \rangle - A(\theta, \Theta) \}$$

where $\langle \Theta, xx^T \rangle := \text{tr}(\Theta xx^T) = \text{tr}(x^T \Theta x) = x^T \Theta x$.

- Correspondence with the original parameters:
 - $\theta = \Sigma^{-1} \mu$ and $\Theta = -\frac{1}{2} \Sigma^{-1}$
 - $A(\theta, \Theta) = \frac{1}{2} (\mu^T \Sigma^{-1} \mu + \log |\Sigma|) = \frac{1}{4} \theta^T \Theta^{-1} \theta - \frac{1}{2} \log |-2\Theta| + \text{const.}$
- Sometimes called Gaussian Markov Random Field (GMRF), esp. when $\Theta_{ij} = 0$ for $(i, j) \notin E$ where E is the edge set of a graph.

Example 29 (Ising model)

- Both a graphical model and an exponential family.
- Used in statistical physics. Allows for complex correlations among discrete variables. Discrete counterpart of GMRF.
- Ingredients:
 - A given graph $G = (V, E)$.
 $V := \{1, \dots, n\}$ vertex set. $E \subset V^2$: edge set.
 - Random variables attached to vertices $X = (X_i : i \in V)$.
 - Each $x_i \in \{-1, +1\}$, the spin of node i .
- Take $\mathcal{X} = \{-1, +1\}^V \simeq \{-1, +1\}^n$.
- $T(X) = (X_i, i \in V, X_i X_j, (i, j) \in E)$.
- Underlying measure is counting (and $h(x) \equiv 1$.)

$$p_{\theta}(x) = \exp \left(\sum_{i \in V} \theta_i x_i + \sum_{(i,j) \in E} \theta_{ij} x_i x_j - A(\theta) \right)$$

Example 30 (Exponential random graph model (ERGM))

- A parametric family of probability distributions on graphs.
- Let $\mathcal{X}$ = space of graphs on n nodes.
- Let $T_i(G)$ be functions on the space of graphs for $i = 1, \dots, k$.
- Usually subgraph counts:

$T_1(G) = \#$ number of edges

$T_2(G) = \#$ number of triangles

$\dots = \dots$

$T_j(G) = \#$ number of r -stars (for given r)

$\dots = \dots$

- Underling measure (counting on graphs)

$$p_{\theta}(G) = \exp \left(\sum_{i=1}^k \theta_i T_i(G) - A(\theta) \right)$$

Focus: full-rank canonical (FRC) exponential families.

Proposition 8

In a canonical exponential family,

- (a) *A is convex on its domain Ω_0 ,*
- (b) *Ω_0 is a convex set.*

Proof. (Enough to show (a). Convexity of Ω_0 follows from convexity of A.)

- Apply Hölder inequality, with $1/p = \alpha$ and $1/q = 1 - \alpha$. (Exercise) Q.E.D.
- Hölder inequality: For $X, Y \geq 0$ a.s.,

$$\mathbb{E}[X^\alpha Y^{1-\alpha}] \leq (\mathbb{E}X)^\alpha (\mathbb{E}Y)^{1-\alpha}, \quad \forall \alpha \in [0, 1]$$

- Expectation can be replaced with integral w.r.t. a general measure:
 $f, g \geq 0$ a.e. $\tilde{\nu}$

$$\int f^\alpha g^{1-\alpha} d\tilde{\nu} \leq \left(\int f d\tilde{\nu} \right)^\alpha \left(\int g d\tilde{\nu} \right)^{1-\alpha}, \quad \forall \alpha \in [0, 1].$$

Proposition 9

In a FRC exponential family, A is C^∞ on $\text{int}(\Omega_0)$ and moreover

$$\mathbb{E}_\theta[T] = \nabla A(\theta), \quad \text{cov}_\theta[T] = \nabla^2 A(\theta)$$

That is,

$$\frac{\partial A}{\partial \theta_i} = \mathbb{E}_\theta[T_i(X)], \quad \frac{\partial^2 A}{\partial \theta_i \partial \theta_j} = \text{cov}_\theta[T_i(X), T_j(X)]$$

Proof sketch.

- Moment generating function (mgf) of T is

$$\begin{aligned} M_T(u) &:= M_T(u; \theta) := \mathbb{E}_\theta[e^{\langle u, T \rangle}] \\ &= \int e^{\langle u, T(x) \rangle} e^{\langle \theta, T(x) \rangle - A(\theta)} d\tilde{\nu}(x) \\ &= e^{A(u+\theta) - A(\theta)} \end{aligned}$$

- If $\theta \in \text{int } \Omega_0$, then M_T is finite in a neighborhood of zero:

$$M_T(u) < \infty, \quad \text{for } \|u\|_2 \leq \varepsilon.$$

- DCT implies M_T is C^∞ in a neighborhood of 0, and we can interchange the order of differentiation and integration.

- Moment generating function:

$$M_T(u) = \mathbb{E}_\theta[e^{\langle u, T \rangle}] = e^{A(u+\theta) - A(\theta)}$$

- We get (fixing θ)

$$M_T(u) \frac{\partial A}{\partial u_i}(u + \theta) = \frac{\partial}{\partial u_i} M_T(u) = \mathbb{E}_\theta \left[\frac{\partial}{\partial u_i} e^{\langle u, T \rangle} \right] = \mathbb{E}_\theta [T_i e^{\langle u, T \rangle}],$$

valid in a neighborhood of 0.

- Evaluating at $u = 0$ gives the result for mean. ($M_T(0) = 1$.)
- Getting the covariance is similar. (Exercise)

Remark 2

- Covariance is positive semidefinite, hence $\nabla^2 A(\theta) \succeq 0$.
- This gives another proof for convexity of A .

Example 31

- $X \sim N(\theta, 1)$:

$$p_{\theta}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \exp\left(\theta x - \frac{1}{2}\theta^2\right)$$

- $A(\theta) = \frac{1}{2}\theta^2$. Hence:

$$\mathbb{E}_{\theta}[X] = A'(\theta) = \theta, \quad \text{var}_{\theta}(X) = A''(\theta) = 1$$

Mean parameters

- Exponential family

$$d\mathbb{P}_\theta(x) = \exp[\langle \theta, T(x) \rangle - A(\theta)] d\nu(x)$$

- Alternative parametrization in terms of **mean parameter** μ :

$$\mu := \mu(\theta) = \mathbb{E}_\theta[T(X)]$$

- Mean parameters are easy to estimate: $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n T(X^{(i)})$.

Example 32 (Two-parameter Gaussian)

- $\mathcal{X} = \mathbb{R}$, $T(x) = (x, x^2)$, $p_\theta(x) = \exp(\theta_1 x + \theta_2 x^2 - A(\theta))$.
- Natural parameter space: $\Omega_0 = \{(\theta_1, \theta_2) : \theta_2 < 0\}$.
- $\theta_1 = \frac{m}{\sigma^2}$ and $\theta_2 = -\frac{1}{2\sigma^2}$ in the original parametrization $N(m, \sigma^2)$.
- Mean parameters:

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} \mathbb{E}[X] \\ \mathbb{E}[X^2] \end{bmatrix} = \begin{bmatrix} m \\ m^2 + \sigma^2 \end{bmatrix} = \begin{bmatrix} -\frac{\theta_1}{2\theta_2} \\ \frac{\theta_1^2}{4\theta_2^2} - \frac{1}{2\theta_2} \end{bmatrix}$$

- Interesting general set:

$$\mathcal{M} := \{\mu \in \mathbb{R}^d \mid \mu = \mathbb{E}_p[T(X)] \text{ for some density } p \text{ w.r.t. } \nu\},$$

the set of realizable mean parameters by **any distribution** (absolutely continuous w.r.t. ν).

- $\mathcal{M}$ is essentially the convex hull of the support of $\nu_{\#}T = \nu \circ T^{-1}$
- More precisely,

$$\text{int}(\mathcal{M}) = \text{int}(\text{co}(\text{supp}(\nu_{\#}T))).$$

$$\mathcal{M} := \{\mu \in \mathbb{R}^d \mid \mu = \mathbb{E}_p[T(X)] \text{ for some density } p \text{ w.r.t. } \nu\},$$

Example 33

- $T(X) = (X, X^2) \in \mathbb{R}^2$ and ν the Lebesgue measure:

$$(\mu_1, \mu_2) = (\mathbb{E}_p[X], \mathbb{E}_p[X^2])$$

- By nonnegativity of the variance we need to have

$$\mathcal{M} \subset \mathcal{M}_0 := \{(\mu_1, \mu_2) : \mu_2 \geq \mu_1^2\}$$

- Any $(\mu_1, \mu_2) \in \text{int } \mathcal{M}_0$ can be realized by a $N(\mu_1, \mu_2 - \mu_1^2)$.
- $\text{bd } \mathcal{M}_0 := \{(\mu_1, \mu_2) : \mu_2 = \mu_1^2\}$ cannot be achieved by a density (why?).
- $\text{bd } \mathcal{M}_0$ can be approached arbitrarily closely by densities.
- We have $\mathcal{M} = \text{int } \mathcal{M}_0$ and $\overline{\mathcal{M}} = \mathcal{M}_0$.

Example 34 (Multivariate Gaussian)

- $T(X) = (X, XX^T)$.
- Let $\mu = \mathbb{E}_p[X]$ and $\Lambda = \mathbb{E}_p[XX^T]$ for some density p w.r.t. Lebesgue measure.
- Covariance matrix is PSD, hence $\Lambda - \mu\mu^T \succeq 0$.
- Closure of $\mathcal{M}$ (set of realizable means) is

$$\overline{\mathcal{M}} := \{(\mu, \Lambda) \mid \Lambda \succeq \mu\mu^T\}$$

- $\text{int}(\overline{\mathcal{M}}) = \mathcal{M} = \{(\mu, \Lambda) \mid \Lambda \succ \mu\mu^T\}$ can be realized by non-degenerate Gaussian distributions $N(\mu, \Lambda - \mu\mu^T)$, a full-rank exponential family.

A remarkable result:

Anything $\in \text{int } \mathcal{M}$ can be realized by an exponential family. Let

$$\Omega := \text{dom } A := \{\theta \in \mathbb{R}^d : A(\theta) < \infty\}$$

Theorem 6

In a **FRC** exponential family, assuming A is **essentially smooth**,

- $\nabla A : \text{int } \Omega \rightarrow \text{int } \mathcal{M}$ is one-to-one and onto.

In other words, ∇A establishes a bijection between $\text{int } \Omega$ and $\text{int } \mathcal{M}$.

- Recall as part of FRC assumption, $\text{int } \Omega \neq \emptyset$.
- WLOG, we can assume $T(x) = x$ (absorb T into the measure, $\nu \circ T^{-1}$),
- That is, we work with the standard family

$$d\mathbb{P}_\theta(x) = \exp(\langle \theta, x \rangle - A(\theta)) d\nu(x)$$

- By Proposition 9, $\mathbb{E}_\theta(X) = \nabla A(\theta)$.
- The proof is a tour de France of convex/real analysis.

Proof sketch

$\nabla A : \text{int } \Omega \rightarrow \text{int } \mathcal{M}$ is **one-to-one (injective)** and onto (surjective)

Let $\Phi := \nabla A$.

1. Φ is regular on $\text{int } \Omega$: $D\Phi = \nabla^2 A \in \mathbb{R}^{d \times d}$ is a full-rank matrix.
True since condition (E2) implies $\nabla^2 A \succ 0$.
2. Φ is injective:
Since condition (E2) implies $\nabla^2 A \succ 0$, we conclude that A is strictly convex. This in turns implies that ∇A is a strictly monotone operator:
 $\langle \nabla A(\theta) - \nabla A(\theta'), \theta - \theta' \rangle > 0$ for $\theta \neq \theta'$.
3. Φ is an open mapping: (maps open sets to open sets) (Corollary 3.27 of ?)
 $U \subset \mathbb{R}^d$ open, $f \in C^1(U, \mathbb{R}^d)$ regular on $U \implies f$ open mapping.
4. By Proposition 9, we have $\nabla A(\text{int } \Omega) \subset \mathcal{M}$.
5. But why $\nabla A(\text{int } \Omega) \subset \text{int } \mathcal{M}$? Follows from ∇A being an open map¹

¹A continuous map is not necessarily open: $x \mapsto \sin(x)$ maps $(0, 4\pi)$ to $[-1, 1]$.

Proof sketch

$\nabla A : \text{int } \Omega \rightarrow \text{int } \mathcal{M}$ is one-to-one (injective) and **onto (surjective)**

It remains to show that $\text{int } \mathcal{M} \subset \nabla A(\text{int } \Omega)$:

For any $\mu \in \text{int } \mathcal{M}$, need to show $\theta \in \text{int } \Omega$ such that $\nabla A(\theta) = \mu$.

6. By applying a shift to ν , WLOG enough to show it for $\mu = 0 \in \text{int } \mathcal{M}$.

7. WTS: $0 \in \text{int } \mathcal{M} \implies \exists \theta \in \text{int } \Omega \text{ s.t. } \nabla A(\theta) = 0.$

In general $0 \notin \text{int } \mathcal{M}$. So, without employing a shift, all the arguments are applied to $\theta \mapsto A(\theta) - \langle \mu, \theta \rangle$.

Proof sketch

$$0 \in \text{int } \mathcal{M} \implies \exists \theta \in \text{int } \Omega \text{ s.t. } \nabla A(\theta) = 0.$$

8. A is lower semi-continuous (lsc) on $\mathbb{R}^d$:

$$\liminf_{\theta \rightarrow \theta_0} A(\theta) \geq A(\theta_0).$$

Follows from Fatou's Lemma.

(lsc only matters at $\text{bd } \Omega$ since A is continuous on $\text{int } \Omega$.)

9. Let $\Gamma_0(\mathbb{R}^d) := \{f : \mathbb{R}^d \rightarrow (-\infty, \infty] \mid f \text{ is proper, convex, lsc}\}$.

(proper means not identically ∞ .)

10. $A \in \Gamma_0(\mathbb{R}^d)$.

11. A is coercive: $\lim_{\|\theta\| \rightarrow \infty} A(\theta) = \infty$.

To be shown.

12. A is essentially smooth:

by assumption.

A function $f \in \Gamma_0(\mathbb{R}^d)$ is *essentially smooth* (a.k.a. steep) if

- (a) f is differentiable on $\text{int dom } f \neq \emptyset$ and
- (b) $\|\nabla f(x_n)\| \rightarrow \infty$ whenever $x_n \rightarrow x \in \text{bd dom } f$.

We in fact only need this for $x \in (\text{dom } f) \cap (\text{bd dom } f)$, parts of the boundary that are in the domain. In particular, (b) is not needed if $\text{dom } f$ is itself open.

$$0 \in \text{int } \mathcal{M} \implies \exists \theta \in \text{int } \Omega \text{ s.t. } \nabla A(\theta) = 0.$$

- 13. A coercive lsc function attains its minimum (over $\mathbb{R}^d$).
- 14. $f \in \Gamma_0(\mathbb{R}^d)$ and essentially smooth $\implies$ minimum cannot be attained at $\text{bd dom } f$.
- 15. If in addition f is strictly convex on $\text{int dom } f$, the minimum is unique.

Lemma 4

Assume that $f \in \Gamma_0(\mathbb{R}^d)$ is coercive, essentially smooth, and strictly convex on $\text{int dom } f$. Then, f attains its unique minimum at some $\bar{x} \in \text{int dom } f$.

A is **coercive**, **essentially smooth**, and **strictly convex on $\text{int dom } A = \text{int } \Omega$**

- 16. Conclude that A attains its minimum at a unique point $\theta \in \text{int } \Omega$.
- 17. Necessary 1st order optimality condition is $\nabla A(\theta) = 0$.
- 18. Done if we show the only remaining piece: coercivity.

A is coercive.

19. For every ε , let $H_{u,\varepsilon} := \{x \in \mathbb{R}^d : \langle x, u \rangle \geq \varepsilon\}$ and let $S = \{u : \|u\|_2 = 1\}$.
20. $0 \in \text{int } \mathcal{M}$ (and full-rank assumption) implies that $\exists \varepsilon > 0$ such that

$$\inf_{u \in S} \nu(H_{u,\varepsilon}) > 0.$$

i.e., $\exists \varepsilon > 0$ and $c \in \mathbb{R}$ such that $\nu(H_{u,\varepsilon}) \geq e^c$ for all $u \in S$.

21. Then, for any $\rho > 0$,

$$\int e^{\langle \rho u, x \rangle} \nu(dx) \geq \int_{H_{u,\varepsilon}} e^{\langle \rho u, x \rangle} \nu(dx) \geq e^{\rho \varepsilon} \nu(H_{u,\varepsilon}) \geq e^{\rho \varepsilon + c}$$

That is, $A(\rho u) \geq \rho \varepsilon + c$.

22. For any $\theta \neq 0$, taking $u = \theta/\|\theta\|$ and $\rho = \|\theta\|$, we obtain

$$A(\theta) \geq \|\theta\| \varepsilon + c, \quad \forall \theta \in \mathbb{R}^d \setminus \{0\}$$

showing that A is coercive.



Side Note: In fact, $\nabla A : \text{int } \Omega \rightarrow \text{int } \mathcal{M}$ is a C^1 diffeomorphism (i.e., a bijection which is C^1 in both directions.)

This follows from Theorem 3.2.8 of ?:

Theorem 7 (Global inverse function theorem)

Let $U \subset \mathbb{R}^d$ be open and $\Phi \in C^1(U, \mathbb{R}^d)$. The following are equivalent:

- $V = \Phi(U)$ is open and $\Phi : U \rightarrow V$ is a C^1 diffeomorphism.*
- Φ is injective and regular on U .*

$\Phi = \nabla A$ is injective and regular on $\text{int } \Omega$ hence a C^1 diffeomorphism.

MLE in exponential family

- Significance of Theorem 6 for statistical inference:
- Assume $X_1, \dots, X_n$ are i.i.d. draws from

$$p_\theta(x) = \exp(\langle \theta, T(x) \rangle - A(\theta)) h(x).$$

- The likelihood is

$$L_X(\theta) = \prod_i p_\theta(X_i) \propto \exp(\langle \theta, \sum_i T(X_i) \rangle - nA(\theta)).$$

- Letting $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n T(X_i)$ the log-likelihood is

$$\ell_X(\theta) = n[\langle \theta, \hat{\mu} \rangle - A(\theta)] + \text{const..}$$

- If $\hat{\mu} \in \text{int } \mathcal{M}$, there exists a unique MLE, solution of $\nabla A(\theta) = \hat{\mu}$.
- That is $\hat{\theta}_{\text{MLE}} = \nabla A^{-1}(\hat{\mu}) = \nabla A^*(\hat{\mu})$.
- A^* is the Fenchel–Legendre conjugate of A .
- If $\hat{\mu} \notin \overline{\mathcal{M}}$, then the MLE does not exist. What happens at the boundary can be determined on a case by case basis.

- A is always lower semi-continuous (lsc).
- If $\Omega = \text{dom } A$ is open, lower semicontinuity implies that $A(\theta) \rightarrow \infty$ as θ approaches the boundary.
(Pick $\theta_0 \in \text{bd } \Omega$, then $\liminf_{\theta \rightarrow \theta_0} A(\theta) \geq A(\theta_0) = \infty$.)
- In other words, if Ω is open, A is automatically essentially smooth.

Example 35 (Two-parameter Gaussian)

- $\mathcal{X} = \mathbb{R}$, $T(x) = (x, x^2)$. $p_\theta(x) = \exp(\theta_1 x + \theta_2 x^2 - A(\theta))$, $\forall x \in \mathcal{X}$.
- $A(\theta) = \log \int e^{\theta_1 x + \theta_2 x^2} dx$. $A(\theta) < \infty$ iff $\theta_2 < 0$.
- Natural parameter space: $\Omega = \text{dom } A = \{(\theta_1, \theta_2) : \theta_2 < 0\}$.
- $\theta_1 = \frac{m}{\sigma^2}$ and $\theta_2 = -\frac{1}{2\sigma^2}$ in original parametrization (m, σ^2) .
- $\mu = \frac{\theta_1}{-2\theta_2}$ and $\sigma^2 = \frac{1}{-2\theta_2}$.

- Mean parametrization:
$$\mu_1 = \frac{\theta_1}{-2\theta_2}, \quad \mu_2 = \left(\frac{\theta_1}{-2\theta_2}\right)^2 + \frac{1}{-2\theta_2}$$
- $A(\theta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2) = \frac{\theta_1^2}{-4\theta_2} + \frac{1}{2} \log \frac{\pi}{-\theta_2}$.
- Easy to verify that $\nabla A(\theta) = (\mu_1, \mu_2)$
and it establishes a bijection between
 $\{(\theta_1, \theta_2) : \theta_2 < 0\} = \text{int } \Omega \leftrightarrow \text{int } \mathcal{M} = \{(\mu_1, \mu_2) : \mu_1^2 > \mu_2\}$

- Note that since A is lsc, $\mu(\theta) = \nabla A(\theta) \rightarrow \infty$ as θ approaches the boundary.
- Show a picture of $\theta \mapsto A(\theta) - \langle \theta, \mu \rangle$ for $\mu = (0, 1)$.

Maximum entropy characterization of exponential family

- Not only exponential families realize any mean, they achieve it with maximum entropy: Solution to

$$\max_p \mathbb{E}_p[-\log p(X)] \quad \text{s.t.} \quad \mathbb{E}_p[T(X)] = \mu,$$

is given by a density of the form $p(x) \propto \exp(\langle \theta, T(x) \rangle)$.

- Discrete case, easy to verify by introducing Lagrange multipliers:
- $\mathcal{X} = \{x_1, \dots, x_K\}$
- $\nu =$ counting meas. and $p_i = p(x_i)$, let $p = (p_1, \dots, p_K)$, and $t_i = T(x_i)$,

$$\begin{aligned} \max_p \quad & -\sum_i p_i \log p_i \\ \text{s.t.} \quad & \sum_i p_i t_i = \mu, \\ & p_i \geq 0, \quad \sum_i p_i = 1 \end{aligned}$$

- Without $\sum_i p_i t_i = \mu$, uniform distribution maximizes the entropy.

Information inequality (Cramer–Rao)

- How small can the variance of an unbiased estimator be? How well can the UMVU do?
- The bound also plays a role in asymptotics.
- Idea: Use **Cauchy–Schwarz (CS)**, also called covariance inequality in this context:

$$(\mathbb{E}XY)^2 \leq (\mathbb{E}X^2)(\mathbb{E}Y^2), \quad \text{or} \quad [\text{cov}(X, Y)]^2 \leq \text{var}(X) \text{var}(Y)$$

- Running assumption: every RV/estimator has finite second moment.
- δ unbiased for some $g(\theta)$, ψ any other estimator

$$\text{var}_\theta(\delta) \geq \frac{[\text{cov}_\theta(\delta, \psi)]^2}{\text{var}_\theta(\psi)} \tag{3}$$

- Need to get rid of δ on the RHS.
- By cleverly choosing ψ , we can obtain good bounds.

- Assume $\mathbb{P}_{\theta+h} \ll \mathbb{P}_{\theta}$: $p_{\theta+h}(x) = 0$ whenever $p_{\theta}(x) = 0$.
- (Local) likelihood ratio is well-defined: (Can define it to be 1 for $0/0$.)

$$L_{\theta,h}(X) = p_{\theta+h}(X)/p_{\theta}(X)$$

- ($= d\mathbb{P}_{\theta+h}/d\mathbb{P}_{\theta}$, Radon-Nikodym derivative of $\mathbb{P}_{\theta+h}$ w.r.t. $\mathbb{P}_{\theta}$.)
- **Change of measure** by integrating against the likelihood ratio:

$$\mathbb{E}_{\theta}[\delta L_{\theta,h}] = \int_{p_{\theta} > 0} \delta L_{\theta,h} p_{\theta} d\mu = \int_{p_{\theta} > 0} \delta p_{\theta+h} d\mu = \mathbb{E}_{\theta+h}[\delta] \quad (4)$$

Note that $p_{\theta+h}$ is concentrated on $\{x : p_{\theta}(x) > 0\}$.

Lemma 5 (Hammersley–Chapman–Robbins (HCR))

Assume $\mathbb{P}_{\theta+h} \ll \mathbb{P}_{\theta}$, and let δ be unbiased for $g(\theta)$. Then,

$$\text{var}_{\theta}(\delta) \geq \frac{[g(\theta+h) - g(\theta)]^2}{\mathbb{E}_{\theta}(L_{\theta,h} - 1)^2}$$

Proof.

- Idea: Apply CS inequality (3) to $\psi = L_{\theta,h} - 1$.
- $\mathbb{E}_{\theta}[\psi] = \mathbb{E}_{\theta}[L_{\theta,h}] - 1 = 0$. (By an application of (4) to $\delta = 1$.)
- Another application of (4) gives²

$$\text{cov}_{\theta}(\delta, \psi) = \mathbb{E}_{\theta}[\delta\psi] = \mathbb{E}_{\theta}[\delta L_{\theta,h}] - \mathbb{E}_{\theta}[\delta] = g(\theta+h) - g(\theta).$$

² ψ is not an unbiased estimator of 0, since it depends on θ . It is not a proper estimator. Not a contradiction with “UMVU is uncorrelated from any unbiased estimator of 0”.

- Assume that θ (and hence h) is a scalar.
- Likelihood ratio approaches 1 as $h \rightarrow 0$:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{1}{h} [L_{\theta, h}(X) - 1] &= \lim_{h \rightarrow 0} \frac{[p_{\theta+h}(X) - p_{\theta}(X)]/h}{p_{\theta}(X)} \\ &= \frac{\partial_{\theta}[p_{\theta}(X)]}{p_{\theta}(X)} = \partial_{\theta}[\log p_{\theta}(X)]\end{aligned}$$

called the **score function**.

- Divide HCR by h , and let $h \rightarrow 0$:

$$\text{var}_{\theta}(\delta) \geq \lim_{h \rightarrow 0} \frac{[g(\theta + h) - g(\theta)]^2/h^2}{\mathbb{E}_{\theta}(L_{\theta, h} - 1)^2/h^2}$$

- Numerator goes to $g'(\theta)$. If justified in exchanging limit and expectation,

$$\text{var}_{\theta}[\delta(X)] \geq \frac{[g'(\theta)]^2}{\mathbb{E}_{\theta}[\partial_{\theta} \log p_{\theta}(X)]^2}$$

Cramer–Rao (formal statement)

- log-likelihood: $\ell_\theta(X) := \log p_\theta(X)$,
- **Score function**: $\dot{\ell}_\theta(X) := \nabla_\theta \ell_\theta(X) = \nabla_\theta \log p_\theta(X) \in \mathbb{R}^d$

Theorem 8 (Cramer–Rao lower bound)

Let $\mathcal{P}$ be dominated family with densities having common support S on some open parameter space $\Omega \subset \mathbb{R}^d$. Assume:

- (a) δ is an unbiased estimator for $g(\theta) \in \mathbb{R}$.
- (b) g is differentiable over Ω , with gradient $\dot{g} = \nabla_\theta g \in \mathbb{R}^d$,
- (c) $\dot{\ell}_\theta(x)$ exists for $x \in S$ and $\theta \in \Omega$,
- (d) At least for $\xi = 1$ and $\xi = \delta$ and $\forall \theta \in \Omega$,

$$\frac{\partial}{\partial \theta_i} \int_S \xi(x) p_\theta(x) d\mu(x) = \int_S \xi(x) \frac{\partial}{\partial \theta_i} p_\theta(x) d\mu(x), \quad \forall i \quad (5)$$

Then,

$$\text{var}_\theta(\delta) \geq \dot{g}(\theta)^T [I(\theta)]^{-1} \dot{g}(\theta)$$

where $I(\theta) = \mathbb{E}_\theta[\dot{\ell}_\theta \dot{\ell}_\theta^T] \in \mathbb{R}^{d \times d}$ is the **Fisher information matrix**.

- Let us rewrite the assumption:

$$\frac{\partial}{\partial \theta_i} \int_S \xi(x) p_\theta(x) d\mu(x) = \int_S \xi(x) \frac{\partial}{\partial \theta_i} p_\theta(x) d\mu(x), \quad \forall i \quad (6)$$

- Note that the right-hand side is:

$$\text{RHS} = \int_S \xi(x) \frac{\partial \log p_\theta(x)}{\partial \theta_i} p_\theta(x) d\mu(x) = \mathbb{E}_\theta(\xi(X) [\dot{\ell}_\theta(X)]_i)$$

- Putting the pieces together

$$\nabla_\theta \mathbb{E}_\theta[\xi] = \mathbb{E}_\theta[\xi \dot{\ell}_\theta] \quad (7)$$

- which is the differential form of the **change of measure** formula:

$$\mathbb{E}_{\theta+h}[\xi] = \mathbb{E}_\theta[\xi L_{\theta,h}]$$

Proof.

- Score function has zero mean, $\mathbb{E}_\theta[\dot{\ell}_\theta] = 0$. (Apply (7) with $\xi = 1$.)
- $\dot{g}(\theta) = \mathbb{E}_\theta[\delta \dot{\ell}_\theta]$ (Apply (7) with $\xi = \delta$.)
- Fix some $a \in \mathbb{R}^d$. Will apply CS inequality (3) with $\psi = a^T \dot{\ell}_\theta$.
- Since $a^T \dot{\ell}_\theta$ is zero mean:

$$a^T \dot{g}(\theta) = \mathbb{E}_\theta[\delta a^T \dot{\ell}_\theta] = \text{cov}_\theta(\delta, a^T \dot{\ell}_\theta).$$

- Similarly,

$$\text{var}_\theta(a^T \dot{\ell}_\theta) = \mathbb{E}_\theta[a^T \dot{\ell}_\theta \dot{\ell}_\theta^T a] = a^T I(\theta) a$$

- CS inequality (3) with $\psi = a^T \dot{\ell}_\theta$ gives:

$$\text{var}_\theta(\delta) \geq \frac{[\text{cov}_\theta(\delta, a^T \dot{\ell}_\theta)]^2}{\text{var}_\theta(a^T \dot{\ell}_\theta)} = \frac{(a^T \dot{g}(\theta))^2}{a^T I(\theta) a}.$$

- Almost done. Problem reduces to (Exercise)

$$\sup_{a \neq 0} \frac{(a^T v)^2}{a^T B a} = v^T B^{-1} v.$$

Hint: Since $B \succ 0$, $B^{-1/2}$ is well-defined; take $z = B^{1/2} a$.

Q.E.D.

- Regularity conditions for interchanging the integral and derivative are key,
- so is the unbiasedness.
- Under same assumptions (recall $\ell_\theta = \log p_\theta(X)$)

$$I(\theta) = \mathbb{E}_\theta[-\ddot{\ell}_\theta] = \mathbb{E}_\theta[-\nabla_\theta^2 \ell_\theta]$$

- $I(\theta)$ measures expected local curvature of the likelihood.
- Attainment of CRB is related to attainment of Cauchy–Schwarz: Wijsman (1973) shows that it happens if and only if we are in the exponential family.
- **Fisher info. is not invariant** to reparametrization:

$$\theta = \theta(\mu) \implies \tilde{I}(\mu) = [\theta'(\mu)]^2 I(\theta)$$

- **CRB is invariant to reparametrization.** (Exercise.)
- Fisher info. is additive over independent sampling.

Multiple parameters

What if $g : \Omega \rightarrow \mathbb{R}^m$ where $\Omega \subset \mathbb{R}^d$?

- Let $J_\theta = (\partial g_i / \partial \theta_j) \in \mathbb{R}^{m \times d}$ be the **Jacobian** of g .
- Then, under similar assumptions (notation: $I_\theta = I(\theta)$):

$$\text{cov}_\theta(\delta) \succeq J_\theta I_\theta^{-1} J_\theta^T$$

for any δ unbiased for $g(\theta)$.

- $A \succeq B$ means $A - B \succeq 0$, i.e., $A - B$ is positive semidefinite (PSD).
- Proof: Fix $u \in \mathbb{R}^m$ and apply the 1-D theorem to $u^T \delta$. (Exercise)

Example 36

- $X_i \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2), i = 1, \dots, n,$
- σ^2 is fixed, $g(\theta) = \theta$.

$$\ell_{\theta}(X) = \log p_{\theta}(X) = \sum_{i=1}^n \log p_{\theta}(X_i) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \theta)^2 + \text{const.}$$

- Differentiating, we get the score function

$$\dot{\ell}_{\theta}(X) = \frac{\partial}{\partial \theta} \log p_{\theta}(X) = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \theta) \implies \ddot{\ell}_{\theta}(X) = -\frac{n}{\sigma^2}.$$

- whence $I(\theta) = n/\sigma^2$.
- CRB is $\text{var}_{\theta}(\delta) \geq \sigma^2/n$ and is achieved by sample mean.

Example 37 (Exponential families)

- $X_i \sim p_\theta(x_i) = h(x_i) \exp(\langle \theta, T(x_i) \rangle - A(\theta)), i = 1, \dots, n.$

$$\ell_\theta(X) = \log p_\theta(X_1, \dots, X_n) = \langle \theta, \sum_i T(X_i) \rangle - nA(\theta) + \text{const.}$$

- whence $I(\theta) = \mathbb{E}_\theta[-\ddot{\ell}_\theta(X)] = n\nabla^2 A(\theta) = n \text{cov}_\theta[T].$

- Consider 1-D case and $n = 1.$
- Want unbiased estimate of the mean parameter: $\mu(\theta) = \mathbb{E}_\theta[T] = A'(\theta).$
- CRB is

$$\frac{[\mu'(\theta)]^2}{I(\theta)} = \frac{[A''(\theta)]^2}{A''(\theta)} = A''(\theta) = \text{var}_\theta(T)$$

i.e., it is attained by $T.$

- General case: $\bar{T} := \frac{1}{n} \sum_{i=1}^n T(X_i)$ attains CRB for the mean parameter:

$$\text{cov}_\theta(\delta) \succeq \text{cov}_\theta(\bar{T}), \quad \forall \delta \text{ s.t. } \mathbb{E}_\theta(\delta) = \mathbb{E}_\theta(\bar{T}).$$

Example 38

- $X_i \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda), i = 1, \dots, n.$
- Exponential family with $T(X) = X$ and mean parameter λ ,
- hence, sample mean $\delta(X) = \frac{1}{n} \sum_i X_i$ achieves the CRB for λ .
- What if we want an unbiased estimate of $g(\lambda) = \lambda^2$?
- Since $I(\lambda) = n/\text{var}_\lambda[X_1] = n/\lambda$ (why?),
- the CRB $= [2\lambda]^2/(n/\lambda) = 4\lambda^3/n$.
- The estimator $T_1 = \frac{1}{n} \sum_{i=1}^n X_i(X_i - 1)$ is unbiased for λ^2 and

$$\text{var}_\lambda(T_1) = 4\lambda^3/n + 2\lambda^2/n > \text{CRB}$$

- $S = \sum_i X_i$ is complete sufficient, hence
- Rao–Blackwellized estimator $T_2 = \mathbb{E}[T_1|S] = S(S-1)/n^2$ is UMVU.
- CRB is not attained since (exercise)

$$\text{var}_\lambda(T_2) = 4\lambda^3/n + 2\lambda^2/n^2 > \text{CRB},$$

A vector of independent Poisson variables, conditioned on their sum has a multinomial distribution. In this case, $\text{Mult}(S, (\frac{1}{n}, \dots, \frac{1}{n}))$.

Average vs. maximum risk optimality

Bayesian Methods:

- Trouble comparing estimators based on whole risk functions $\theta \mapsto R(\theta, \delta)$.
- The **Bayesian approach**: reduce to (weighted) average risk.
- Assumes that the parameter is a random variable Θ with some distribution Λ , called the **prior**, having density $\pi(\theta)$ (w.r.t. to say Lebesgue).
- Choice of the prior is important in the Bayesian framework.
- Frequentest perspective: Bayes estimators have desirable properties.

- Recall the decision-theoretic framework:
- Family of distributions $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Omega\}$.
- Bayesian framework:
interpret $\mathbb{P}_\theta$ as conditional distribution of X given $\Theta = \theta$,

$$\mathbb{P}_\theta(A) = \mathbb{P}(X \in A \mid \Theta = \theta)$$

- Together with the marginal (**prior**) distribution of Θ , we have the joint distribution of (Θ, X) .
- Recall the risk defined as

$$R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))] = \mathbb{E}[L(\theta, \delta(X)) \mid \Theta = \theta]$$

or in other words, $R(\Theta, \delta) = \mathbb{E}[L(\Theta, \delta(X)) \mid \Theta]$.

- The **Bayes risk** is

$$r(\Lambda, \delta) = \mathbb{E}[R(\Theta, \delta)] = \mathbb{E}[L(\Theta, \delta(X))].$$

- Write $p(x|\theta) = p_\theta(x)$ for density of $\mathbb{P}_\theta$.
- Recall that

$$R(\theta, \delta) = \int L(\theta, \delta(x))p(x|\theta)dx.$$

Then,

$$r(\Lambda, \delta) = \int \pi(\theta)R(\theta, \delta)d\theta = \int \pi(\theta) \left[\int L(\theta, \delta(x))p(x|\theta)dx \right] d\theta.$$

- We rarely used this explicit form.

- A **Bayes rule** or estimator w.r.t. Λ , denoted as δ_Λ , is a minimizer of the Bayes risk:

$$r(\Lambda, \delta_\Lambda) = \min_{\delta} r(\Lambda, \delta)$$

- Depends both on the prior Λ and the loss L .

Theorem 9 (Existence of Bayes estimators)

Assume that

- (a) $\exists \delta'$ with $r(\Lambda, \delta') < \infty$
- (b) **Posterior risk** has a minimizer for μ -almost all x , that is,

$$\delta_\Lambda(x) := \operatorname{argmin}_{a \in \mathcal{A}} \mathbb{E}[L(\Theta, a) | X = x]$$

is well-defined for μ -almost all x . (Measurable selection.)

Then, δ_Λ is a Bayes rule.

Proof. Condition (a) is to guarantee that we can use Fubini theorem.

- By definition of δ_Λ , for any δ we have $\mathbb{E}[L(\Theta, \delta) | X] \geq \mathbb{E}[L(\Theta, \delta_\Lambda) | X]$.
- Taking expectation and using smoothing finishes the proof.

- Posterior risk can be computed based on the **posterior distribution** of Θ given $X = x$. Bayes rule gives

$$\pi(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{m(x)} \propto p(x|\theta)\pi(\theta)$$

where $m(x) = \int \pi(\theta)p(x|\theta)d\theta$ is the marginal distribution of X .

- Posterior is proportional to prior times the likelihood.

Example 39

Bayes estimators for two simple loss functions:

- Quadratic (or ℓ_2) loss: $L(\theta, a) = (g(\theta) - a)^2$:

$$\delta_\Lambda(x) = \min_a \mathbb{E}[(g(\Theta) - a)^2 | X = x] = \mathbb{E}[g(\Theta) | X = x].$$

For $g(\theta) = \theta$ reduces to the posterior mean.

- ℓ_1 loss: $L(\theta, a) = |\theta - a|$: Here, $\delta_\Lambda(x) = \text{median}(\Theta | X = x)$ is one possible Bayes estimator. (Not unique in this case.)

Example 40 (Binomial)

- $X \sim \text{Bin}(n, \theta)$.
- PMF is $p(x|\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$.
- Put a Beta prior on Θ , with *hyperparameters* $\alpha, \beta > 0$,

$$\pi(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1} \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

- We have $\pi(\theta|x) \propto p_\theta(x)\pi(\theta) \propto \theta^{x+\alpha-1} (1 - \theta)^{n-x+\beta-1}$
- showing that $\Theta|X = x \sim \text{Beta}(\alpha + x, n - x + \beta)$, whence

$$\delta_\lambda(x) := \mathbb{E}[\Theta|X = x] = \frac{x + \alpha}{n + \alpha + \beta} = (1 - \lambda) \frac{x}{n} + \lambda \frac{\alpha}{\alpha + \beta}$$

where $\lambda = \frac{\alpha + \beta}{n + \alpha + \beta}$.

- Note: $\alpha/(\alpha + \beta)$ is the prior mean, and x/n is the MLE (or unbiased estimator of the mean parameter).
- No coincidence, happens in a general exponential family.

Example 41 (Normal location family)

- Assume that $X_i|\Theta = \theta \sim N(\theta, \sigma^2)$.
- Put a Gaussian prior on $\Theta \sim N(\mu, b^2)$.
- The model is equivalent to

$$X_i = \Theta + w_i, \quad w_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad \text{for } i = 1, \dots, n$$

- Reparametrize in terms of precisions $\tau^2 = 1/b^2$ and $\gamma^2 = 1/\sigma^2$.
- $(\Theta, X_1, \dots, X_n)$ is jointly Gaussian, and the posterior is

$$\Theta|X = x \sim N\left(\underbrace{(1 - \lambda_n)\bar{x} + \lambda_n\mu}_{\delta_\Lambda(x)}, 1/\tilde{\tau}_n^2\right)$$

where

$$\bar{x} = \frac{1}{n} \sum x_i, \quad \tilde{\tau}_n^2 = n\gamma^2 + \tau^2, \quad \lambda_n = \tau^2/\tilde{\tau}_n^2 \in [0, 1]$$

- Continued ...

- With

$$\bar{x} = \frac{1}{n} \sum x_i, \quad \tilde{\tau}_n^2 = n\gamma^2 + \tau^2, \quad \lambda_n = \tau^2 / \tilde{\tau}_n^2 \in [0, 1]$$

- We have

$$\Theta|X=x \sim N(\delta_\Lambda(x), 1/\tilde{\tau}_n^2)$$

- Posterior mean $\delta_\Lambda(x)$, i.e., the Bayes rule for ℓ_2 loss, is

$$\delta_\Lambda(x) := (1 - \lambda_n)\bar{x} + \lambda_n\mu$$

which is a convex combination of $\bar{x}$ and μ and we have

- $\delta_\Lambda(x) \rightarrow \bar{x}$ if $n \rightarrow \infty$ or $\text{SNR} = \gamma^2/\tau^2 \rightarrow \infty$.
- $\delta_\Lambda(x) \rightarrow \mu$ if $\text{SNR} = \gamma^2/\tau^2 \rightarrow 0$.

Conjugate priors

- The two examples above are examples of *conjugacy*.
- A family $\mathcal{Q} = \{\pi(\cdot)\}$ of priors is conjugate to a family of likelihoods $\mathcal{P} = \{p(\cdot | \theta)\}$ if the corresponding posteriors also belong to $\mathcal{Q}$.
- Example of conjugate families

$\mathcal{Q}$	normal	beta	Dirichlet
$\mathcal{P}$	normal	binomial	multinomial

Example 42 (Exponential families)

We have the following conjugate pairs

$$p(x|\theta) = \exp \{ \langle \eta(\theta), T(x) \rangle - A(\theta) \}$$

$$q_{a,b}(\theta) = \exp \{ \langle a, \eta(\theta) \rangle + bA(\theta) - B(a, b) \}$$

Example 43 (Improper priors)

- $X_i \sim N(\theta, \sigma^2), i = 1, \dots, n.$
- Is $\delta(x) = \frac{1}{n} \sum x_i$, a Bayes estimator w.r.t. some prior?
- Not if we require proper priors (finite measures): $\int \pi(\theta) d\theta < \infty$, in which case π can be normalized to integrate to 1.
- Need a uniform (proper) prior on the whole $\mathbb{R}$ which does not exist.
- An improper prior can still be used if the posterior is well-defined. (Generalized Bayes.)
- Alternatively, $\delta(x)$ is the limit of Bayes rules for a sequence of proper priors. (see also the Beta-Binomial example.)

Comment on the uniqueness of the Bayes estimator.

Theorem 10 (TPE 4.1.4)

Let Q be the marginal distribution of X , that is, $Q(A) = \int \mathbb{P}_\theta(X \in A) d\Lambda(\theta)$. Recall that δ_Λ is (a) Bayes estimator. Assume that

- The loss function is strictly convex,
- $r(\Lambda, \delta_\Lambda) < \infty$,
- $Q \text{ a.e.} \implies \mathcal{P} \text{ a.e.}$. Equivalently, $\mathbb{P}_\theta \ll Q$ for all $\theta \in \Omega$.

Then, there is a unique Bayes estimator.

Minimax criterion

- Instead of averaging the risk: look at the worst-case or **maximum risk**:

$$\overline{R}(\delta) := \sup_{\theta \in \Omega} R(\theta, \delta)$$

- More in accord with an adversarial nature. (A zero-sum game.)

Definition 13

An estimator δ_* is minimax if $\min_{\delta \in \mathcal{D}} \overline{R}(\delta) = \overline{R}(\delta_*)$.

- An effective strategy for finding minimax estimators is to look among the Bayes estimators:
- The minimax problem is: $\inf_{\delta} \sup_{\theta} R(\theta, \delta)$.
- We generalize this to: $\inf_{\delta} \sup_{\Lambda} r(\Lambda, \delta)$

- Recall: δ_Λ a Bayes estimator for prior Λ , and **Bayes risk**,

$$r_\Lambda = \inf_{\delta} r(\Lambda, \delta) = r(\Lambda, \delta_\Lambda),$$

(Last equality: assume r_Λ is finite and is achieved.)

- Can order priors based on their Bayes risk:

Definition 14

Λ_* is a least favorable prior if $r_{\Lambda_*} \geq r_\Lambda$ for any prior Λ .

- For a least favorable prior, we have

$$r_{\Lambda_*} = \sup_{\Lambda} r_\Lambda = \sup_{\Lambda} \inf_{\delta} r(\Lambda, \delta) \leq \inf_{\delta} \sup_{\Lambda} r(\Lambda, \delta) =: \inf_{\delta} \bar{r}(\delta)$$

where $\bar{r}(\delta) = \sup_{\Lambda} r(\Lambda, \delta)$ is a generalization of the maximum risk $\bar{R}(\delta)$.

- Interested in situations where equality holds.

Characterization of minimax estimators

Theorem 11 (TPE 5.1.4)

Assume that δ_Λ is Bayes for Λ , and $r(\Lambda, \delta_\Lambda) = \overline{R}(\delta_\Lambda)$. Then,

- δ_Λ is minimax.
- Λ is least favorable.
- If δ_Λ is the unique Bayes estimator (a.e. $\mathcal{P}$), then it is the unique minimax estimator.

Proof of minimaxity of δ_Λ :

- Maximum risk is **always** lower-bounded by Bayes risk,

$$\overline{R}(\delta) = \sup_{\theta \in \Omega} R(\theta, \delta) \geq \int R(\theta, \delta) d\Lambda(\theta) = r(\Lambda, \delta), \quad \forall \delta$$

- $\overline{R}(\delta) \geq r(\Lambda, \delta) \geq r_\Lambda = \overline{R}(\delta_\Lambda)$. (Last equality by assumption.)

Rest of the proof:

- $\overline{R}(\delta) \geq r(\Lambda, \delta) \geq r_\Lambda = \overline{R}(\delta_\Lambda)$. (Last equality by assumption.)
- Uniqueness of the Bayes rule makes **second inequality** strict for $\delta \neq \delta_\Lambda$, showing the uniqueness of minimax rule.
- On the other hand,

$$r_{\Lambda'} \leq r(\Lambda', \delta_\Lambda) \leq \overline{R}(\delta_\Lambda) = r_\Lambda.$$

showing that Λ is least favorable.

- A decision rule δ is called an equalizer if it has constant risk:

$$R(\theta', \delta) = R(\theta, \delta), \quad \text{for all } \theta, \theta' \in \Omega.$$

- Let $\omega(\delta) := \{\theta : R(\theta, \delta) = \bar{R}(\delta)\} = \operatorname{argmax}_{\theta} R(\theta, \delta)$.
- (δ is an equalizer if $\omega(\delta) = \Omega$.)

Corollary 4 (TPE 5.1.5–6)

- (a) *A Bayes estimator with constant risk (i.e., an equalizer) is minimax.*
- (b) *A Bayes estimator δ_{Λ} is minimax if $\Lambda(\omega(\delta_{\Lambda})) = 1$.*

- Both of these conditions are sufficient, not necessary.
- (b) is weaker than (a).
- Strategy: Find a prior Λ whose support is contained in $\operatorname{argmax}_{\theta} R(\theta, \delta_{\Lambda})$.

Example 44 (Bernoulli, continuous parameter space)

- $X \sim \text{Ber}(\theta)$ with quadratic loss, and $\Theta \in [0, 1]$.
- Given a prior Λ on $[0, 1]$, let $m_1 = \mathbb{E}[\Theta]$ and $m_2 = \mathbb{E}[\Theta^2]$.
- Frequentist risk

$$\begin{aligned} R(\theta, \delta) &= (\delta_0 - \theta)^2(1 - \theta) + (\delta_1 - \theta)^2\theta \\ &= \theta^2[1 + 2(\delta_0 - \delta_1)] + \theta(\delta_1^2 - \delta_0^2 - 2\delta_0) + \delta_0^2 \end{aligned}$$

- Bayes risk

$$r(\Lambda, \delta) = \mathbb{E}[R(\Theta, \delta)] = m_2[1 + 2(\delta_0 - \delta_1)] + m_1(\delta_1^2 - \delta_0^2 - 2\delta_0) + \delta_0^2$$

- Bayes decision rule is found by minimizing $r(\Lambda, \delta)$ w.r.t. δ_0, δ_1 ,

$$\delta_1^* = \frac{m_2}{m_1}, \quad \delta_0^* = \frac{m_1 - m_2}{1 - m_1}.$$

- Bayes decision rule is found by minimizing $r(\Lambda, \delta)$ w.r.t. δ_0, δ_1 ,

$$\delta_1^* = \frac{m_2}{m_1}, \quad \delta_0^* = \frac{m_1 - m_2}{1 - m_1}.$$

- Aside: Since $p(\theta|x) = C\theta^x(1-\theta)^{1-x}\pi(\theta)$, check that $\delta_x^* = \mathbb{E}[\Theta|X=x]$, $x=0,1$ as it should be:

$$\delta_x^* = \frac{\int \theta^{x+1}(1-\theta)^{1-x}\pi(\theta)d\theta}{\int \theta^x(1-\theta)^{1-x}\pi(\theta)d\theta}.$$

- A general rule δ is an equalizer, i.e., $R(\theta, \delta)$ does not depend on θ , if

$$\delta_1 - \delta_0 = 1/2 \text{ and } \delta_1^2 - \delta_0^2 - 2\delta_0 = 0.$$

- These equations have a single solution: $\delta_0 = 1/4$ and $\delta_1 = 3/4$.
(There is a unique equalizer rule.)
- Equalizer Bayes rule: need $3/4 = \frac{m_2}{m_1}$, and $1/4 = \frac{m_1 - m_2}{1 - m_1}$.
- Solving: $m_1^* = 1/2$ and $m_2^* = 3/8$.
- Need prior Λ that has these moments; $\Lambda = \text{Beta}(1/2, 1/2)$ fits the bill.
- This is a **least favorable prior**.
- Corresponding Bayes, hence minimax, risk is $1/16$.

- The above can be generalized to an i.i.d. sample of size n :
 $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, where $\text{Beta}(n/2, n/2)$ is least favorable and the associated minimax risk is $\frac{1}{4(\sqrt{n}+1)^2}$.
- Compare with the risk of the sample mean $R(\theta, \bar{X}) = \theta(1 - \theta)/n$.

Example 45 (Bernoulli, discrete parameter space)

- Let $X \sim \text{Ber}(\theta)$ and $\Omega = \{1/3, 2/3\} =: \{a, b\}$.
- Take $L(\theta, \delta) = (\theta - \delta)^2$.
- Any (nonrandomized) decision rule is specified by a pair of numbers (δ_0, δ_1) .
- Any prior π specified by a single number $\pi_a = \mathbb{P}(\Theta = a) \in [0, 1]$.
- Frequentist risk

$$R(\theta, \delta) = (\delta_0 - \theta)^2(1 - \theta) + (\delta_1 - \theta)^2\theta$$

- Bayes risk $r(\pi, \delta) = \mathbb{E}[R(\Theta, \delta)]$,

$$r(\pi, \delta) = \pi_a R(a, \delta) + (1 - \pi_a) R(b, \delta)$$

- Take derivatives w.r.t. δ_0, δ_1 , set to zero, find the Bayes rule

$$\delta_0^* = \frac{a\pi_a(1-a) + b(1-b)(1-\pi_a)}{(1-a)\pi_a + (1-b)(1-\pi_a)}$$

$$\delta_1^* = \frac{a^2\pi_a + b^2(1-\pi_a)}{a\pi_a + b(1-\pi_a)}$$

- For $a = 1/3 = 1 - b$,

$$\delta_0^* = \frac{2}{3(\pi_a + 1)} \quad \text{and} \quad \delta_1^* = \frac{4 - 3\pi_a}{6 - 3\pi_a}.$$

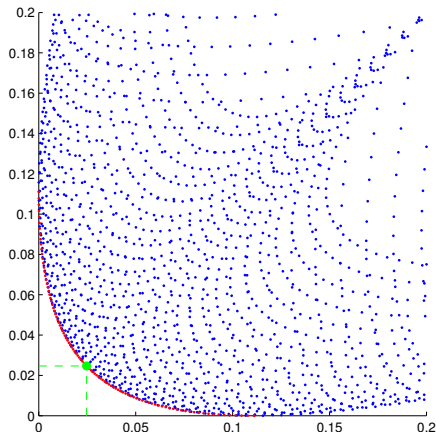
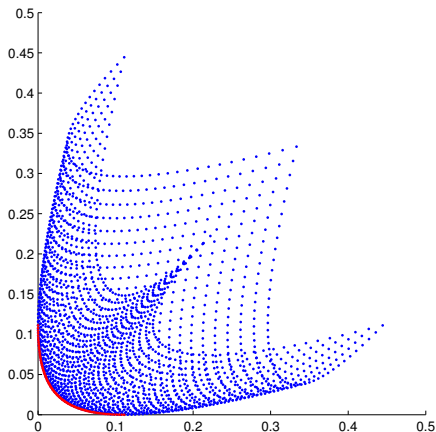
- Equalizer rule, one that $R(a, \delta) = R(b, \delta)$:

$$(a+b)[2(\delta_0 - \delta_1) + 1] + \delta_1^2 - \delta_0^2 - 2\delta_0 = 0.$$

- A Bayes rule that is also equalizer occurs for $\pi_a^* = 1/2$.
- This is the least favorable prior.
- Corrsponding rule $(\delta_0^*, \delta_1^*) = (4/9, 5/9)$ is minimax.

Geometry of Bayes and Minimax

- Risk body for two-parameter Bernoulli $\Omega = \{1/3, 2/3\}$.
- Deterministic rules, Bayes rules, minimax rule.



Geometry of Bayes and minimax for finite Ω

- Assume $\Omega = \{\theta_1, \dots, \theta_k\}$ finite, and consider the risk set (or body)

$$S = \{(y_1, \dots, y_k) \mid y_i = R(\theta_i, \delta) \text{ for some } \delta\} \subset \mathbb{R}^k.$$

- Alternatively, define $\rho : \mathcal{D} \rightarrow \mathbb{R}^k$ by

$$\rho(\delta) = (R(\theta_1, \delta), \dots, R(\theta_k, \delta))$$

where $\mathcal{D}$ is the set of **randomized** decision rules.

- S is the image of $\mathcal{D}$ under ρ , i.e., $S = \rho(\mathcal{D})$.

Lemma 6

S is a convex set (with randomized estimators).

Proof. For $\delta, \delta' \in \mathcal{D}$, and $a \in [0, 1]$, we can form a randomized decision rule δ_a such that $S \ni R(\theta, \delta_a) = aR(\theta, \delta) + (1 - a)R(\theta, \delta')$. (Exercise.)

- Every prior Λ corresponds to a vector $\lambda = (\lambda_1, \dots, \lambda_k) \in \mathbb{R}^k$, via $\Lambda(\{\theta_i\}) = \lambda_i$. Note that λ lies in the **(k - 1)-simplex**,

$$\Delta := \{(\lambda_1, \dots, \lambda_k) \in \mathbb{R}_+^k : \sum_{i=1}^k \lambda_i = 1\}$$

- Bayes risk is

$$r(\Lambda, \delta) = \mathbb{E}[R(\Theta, \delta)] = \sum_{i=1}^k \lambda_i R(\theta_i, \delta) = \lambda^T \rho(\delta)$$

- Hence finding the Bayes rule is equivalent to

$$\inf_{\delta \in \mathcal{D}} r(\Lambda, \delta) = \inf_{\delta \in \mathcal{D}} \lambda^T \rho(\delta) = \inf_{y \in S} \lambda^T y$$

a **convex problem in $\mathbb{R}^k$** . **Minimax problem** is

$$\inf_{\delta \in \mathcal{D}} \|\rho(\delta)\|_\infty = \inf_{y \in S} \|y\|_\infty$$

Finding the least favorable prior corresponds to $\sup_{\lambda \in \Delta} [\inf_{y \in S} \lambda^T y]$.

Admissibility of Bayes rules

- In general, a unique (a.e. $\mathcal{P}$) Bayes rule is admissible (TPE 5.2.4).
- Complete answer to admissibility question for finite parameter spaces.

Proposition 10

*Assume $\Omega = \{\theta_1, \dots, \theta_k\}$ and that δ_λ is Bayes rule for λ .
If $\lambda_i > 0$ for all i , then δ_λ is admissible.*

Proof. If δ_λ is inadmissible, there is δ such that

$$R(\theta_i, \delta) \leq R(\theta_i, \delta_\lambda), \forall i$$

with strict inequality for some j . Then,

$$\sum_i \lambda_i R(\theta_i, \delta) < \sum_i \lambda_i R(\theta_i, \delta_\lambda).$$

Q.E.D.

Proposition 11

Assume $\Omega = \{\theta_1, \dots, \theta_k\}$ and δ admissible. Then δ is Bayes w.r.t. some prior λ .

Proof.

- Let $x := \rho(\delta)$, the risk vector of δ and $Q_x := \{y \in \mathbb{R}^k \mid y_i \leq x_i\} \setminus \{x\}$.
- Q_x is convex.
(Removing an extreme point from a convex set preserves convexity.)
- Admissibility means $Q_x \cap S = \emptyset$.
- Two non-empty disjoint convex sets $\subset \mathbb{R}^k$, can be separated by a hyperplane:

$$\exists u \neq 0 \text{ s.t. } u^T z \leq u^T y, \text{ for all } z \in Q_x \text{ and } y \in S.$$

- Suppose we can choose u to have nonnegative coordinates.
(Proof by contradiction. (Exercise.))

- Since $u \neq 0$, we can set $\lambda = u/(\sum_i u_i) \in \Delta^{k-1}$.
- $\lambda^T z \leq \inf_{y \in S} \lambda^T y = r_\lambda, \quad \forall z \in Q_x$.
- Taking $\{z_n\} \subset Q_x$ such that $z_n \rightarrow x$, we obtain $\lambda^T x \leq r_\lambda$.
- But, by definition of optimal Bayes risk $r_\lambda \leq \lambda^T x$, hence $r_\lambda = \lambda^T x$.

- Setup: An i.i.d. sample of size n from a model $\mathcal{P}^{(1)} = \{\mathbb{P}_\theta : \theta \in \Omega\}$ on sample space $\mathcal{X}$, i.e.,

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{P}_\theta.$$

- Full model is actually $\mathcal{P}^{(n)} = \{\mathbb{P}_\theta^{\otimes n} : \theta \in \Omega\}$ and sample space $\mathcal{X}^n$.
- M -estimators: those obtained as solutions of optimization problems.

Definition 15

Given a family of functions $m_\theta : \mathcal{X} \rightarrow \mathbb{R}$, for $\theta \in \Omega$, the corresponding M -estimator based on $X_1, \dots, X_n$ is

$$\hat{\theta}_n := \hat{\theta}_n(X_1, \dots, X_n) := \operatorname{argmax}_{\theta \in \Omega} \frac{1}{n} \sum_{i=1}^n m_\theta(X_i)$$

- Often $M_n(\theta) := \frac{1}{n} \sum_{i=1}^n m_\theta(X_i)$, a random function.

- Alternative approach is to specify $\hat{\theta}$ as **Z-estimator**, i.e., the solution of a set of **estimating equations**

$$\Psi_n(\theta) := \frac{1}{n} \sum_{i=1}^n \psi(X_i, \theta) = 0.$$

- Often 1st-order optimality conditions for an M -estimator produce a set of estimating equations.
(Simplistic in general, ignoring the possibility of constraints imposed by Ω .)

Example 46

1. $m_\theta(x) = -(x - \theta)^2$, then $M_n(\theta) = -\frac{1}{n} \sum_{i=1}^n (X_i - \theta)^2$, giving $\hat{\theta} = \bar{X}$.
 2. $m_\theta(x) = -|x - \theta|$, then $M_n(\theta) = -\frac{1}{n} \sum_{i=1}^n |X_i - \theta|$, giving $\hat{\theta} = \text{median}(X_1, \dots, X_n)$.
 3. $m_\theta(x) = \log p_\theta(x)$, then $M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log p_\theta(X_i)$, giving the maximum likelihood estimator (MLE).
- In location family with $p_\theta(x) = C \exp(-\beta|x - \theta|^p)$, MLE is equivalent to an M -estimator with $m_\theta(x) = -|x - \theta|^p$.
 - $p = 2$, Gaussian distribution. (Case 1)
 - $p = 1$, Laplace distribution. (Case 2)
 - Corresponding Z -estimator forms of 1. and 2. are obtained for $\psi_\theta(x) = x - \theta$ and $\psi_\theta(x) = \text{sign}(x - \theta)$, obtained by differentiation (or sub-differentiation) of m_θ .

Example 47 (Method of Moments (MOM))

- Find θ by matching empirical and population (true) moments:

$$\mathbb{E}_{\theta}[X_1^k] = \frac{1}{n} \sum_{i=1}^n X_i^k, \quad k = 1, 2, \dots, d.$$

- Usually d is the dimension of parameter θ (d equations in d unknown).
- A set of estimating equations for $\psi_{\theta}(x) = x^k - \mathbb{E}_{\theta}[X_1^k]$.
- A generalized version of MOM is to solve

$$\mathbb{E}_{\theta}[\varphi_k(X_1)] = \frac{1}{n} \sum_{i=1}^n \varphi_k(X_i), \quad k = 1, 2, \dots, d.$$

for some collection of function $\{\varphi_k\}$, corresponding to Z-estimator with $\psi_{\theta}(x) = \varphi_k(x) - \mathbb{E}_{\theta}(\varphi(X_1))$.

- In canonical exponential families:

$$X_i \stackrel{\text{iid}}{\sim} p_\theta(x) \propto \exp\{\langle \theta, T(x) \rangle - A(\theta)\}$$

ML and MOM are equivalent.

- MLE is the M -estimator associated with

$$m_\theta(x) = \log p_\theta(x) = \langle \theta, T(x) \rangle - A(\theta)$$

hence

$$M_n(\theta) = \frac{1}{n} \sum_i [\langle \theta, T(X_i) \rangle - A(\theta)] = \langle \theta, \hat{T} \rangle - A(\theta)$$

where $\hat{T} = \frac{1}{n} \sum_i T(X_i)$ is the empirical mean of the sufficient statistic.

- MLE is

$$\hat{\theta}_{\text{mle}} = \operatorname{argmax}_{\theta \in \Omega} \{ \langle \theta, \hat{T} \rangle - A(\theta) \}$$

- Setting derivatives to zero gives $\hat{T} = \nabla A(\hat{\theta}_{\text{mle}})$. (First-order optimality.)
- Since $\mathbb{E}_\theta[T] = \nabla A(\theta)$, MLE is the solution of

$$\mathbb{E}_\theta[T] = \hat{T}$$

for θ , which is a MOM estimator. (If you will $\mathbb{E}_\theta[T(X_1)] = \frac{1}{n} \sum_i T(X_i)$.)

- Recall that $\mu = \nabla A(\theta)$ is the mean parameterization.
- The inverse of this map is $\theta = \nabla A^*(\mu)$ where

$$A^*(\mu) = \sup_{\theta \in \Omega} \{ \langle \theta, \mu \rangle - A(\theta) \}$$

is the conjugate dual of A . (Exercise.)

- So $\hat{\theta}_{\text{mle}} = \nabla A^*(\hat{T})$, assuming that $\hat{T} \in \text{int}(\text{dom}(A^*))$.

Asymptotics or large-sample theory

Zeroth-order (consistency)

- Statistical behavior of estimators, in particular M -estimators, as $n \rightarrow \infty$.
- For concreteness consider the sequence:

$$\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n) = \operatorname{argmax}_{\theta \in \Omega} \frac{1}{n} \sum_{i=1}^n m_{\theta}(X_i)$$

Definition 16

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{P}_{\theta_0}$. We say that $\{\hat{\theta}_n\}$ is **consistent** if $\hat{\theta}_n \xrightarrow{P} \theta_0$.

- Equivalently,

$$\forall \varepsilon > 0, \quad \mathbb{P}(d(\hat{\theta}_n, \theta_0) > \varepsilon) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

- Usually $d(\hat{\theta}_n, \theta_0) = \|\hat{\theta}_n - \theta_0\|$ for Euclidian parameters spaces $\Omega \subset \mathbb{R}^d$.
- For $d = 1$, $d(\hat{\theta}_n, \theta_0) = |\hat{\theta}_n - \theta_0|$.

- We write $Z_n = o_p(1)$ if $Z_n \xrightarrow{P} 0$.
- By the WLLN, for any fixed θ , we have (assuming $\mathbb{E}_{\theta_0}|m_{\theta}(X_1)| < \infty$)

$$\frac{1}{n} \sum_{i=1}^n m_{\theta}(X_i) \xrightarrow{P} \mathbb{E}_{\theta_0}[m_{\theta}(X_1)]$$

- Letting $M(\theta) := \mathbb{E}_{\theta_0}[m_{\theta}(X_1)]$, for any fixed θ , $M_n(\theta) \xrightarrow{P} M(\theta)$.
- If θ_0 is the maximum of M over Ω , hope $\hat{\theta}_n$ which is the maximum of M_n over Ω approaches it.
- However, pointwise convergence of M_n to M is not enough; need **uniform convergence**, i.e.,

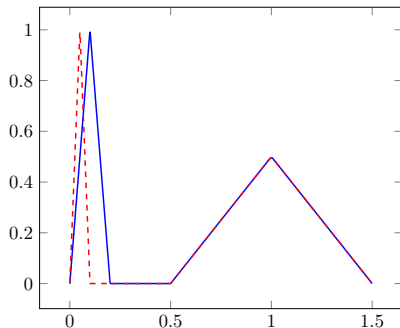
$$\|M_n - M\|_{\infty} := \sup_{\theta \in \Omega} |M_n(\theta) - M(\theta)|$$

to go to zero in probability.

Why uniform convergence?

- Even a nonrandom example is enough:
- Here $M_n(t) \rightarrow M(t)$ pointwise, but $M_n(t_0) = 1$ and $M(t_0) = 1/2$.

$$M_n(t) = \begin{cases} 1 - n|x - \frac{1}{n}| & |x| < \frac{2}{n} \\ \frac{1}{2} - |x - 1| & \frac{1}{2} < |x| < \frac{3}{2} \\ 0 & \text{otherwise} \end{cases}$$
$$M(t) = \begin{cases} \frac{1}{2} - |x - 1| & \frac{1}{2} < |x| < \frac{3}{2} \\ 0 & \text{otherwise} \end{cases}$$



Theorem 12 (AS 5.7 modified)

Let M_n be random functions, and let M be a fixed function of θ . Let

$$\hat{\theta}_n \in \operatorname{argmax}_{\theta \in \Omega} M_n(\theta) \quad (\text{cond-M})$$

be well-defined. Assume:

- (a) $\|M_n - M\|_\infty \xrightarrow{P} 0$. (Uniform convergence.)
- (b) $(\forall \varepsilon > 0) \sup_{\theta: d(\theta, \theta_0) \geq \varepsilon} M(\theta) < M(\theta_0)$. (M has well-separated maxima.)

Then $\hat{\theta}_n$ is consistent, i.e., $\hat{\theta}_n \xrightarrow{P} \theta_0$.

- By optimality of $\hat{\theta}$ for M_n , we have $M_n(\theta_0) \leq M_n(\hat{\theta}_n)$, or

$$0 \leq M_n(\hat{\theta}_n) - M_n(\theta_0) \quad \text{Basic inequality}$$

- By adding and subtracting, we get

$$\begin{aligned} M(\theta_0) - M(\hat{\theta}_n) &\leq M_n(\hat{\theta}_n) - M(\hat{\theta}_n) - [M_n(\theta_0) - M(\theta_0)] \\ &\leq 2\|M_n - M\|_\infty \end{aligned}$$

(We are keeping random deviations from the mean on one side and fixed functions on the other side.)

- Fix some $\varepsilon > 0$, and let

$$\eta(\varepsilon) := M(\theta_0) - \sup_{d(\theta, \theta_0) \geq \varepsilon} M(\theta) = \inf_{d(\theta, \theta_0) \geq \varepsilon} [M(\theta_0) - M(\theta)]$$

- By assumption (b) $\eta(\varepsilon) > 0$.
- Since $d(\theta, \theta_0) \geq \varepsilon$ implies $M(\theta_0) - M(\theta) \geq \eta(\varepsilon)$, we have

$$\begin{aligned} \mathbb{P}(d(\hat{\theta}_n, \theta_0) \geq \varepsilon) &\leq \mathbb{P}(M(\theta_0) - M(\hat{\theta}_n) \geq \eta(\varepsilon)) \\ &\leq \mathbb{P}(2\|M_n - M\|_\infty \geq \eta(\varepsilon)) \rightarrow 0 \end{aligned}$$

by assumption (a).

Q.E.D.

Remark 3

A key step is bounding $M_n(\hat{\theta}_n) - M(\hat{\theta}_n)$ with $\|M_n - M\|_\infty$.

Exercise: Condition (cond-M) can be replaced with $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_p(1)$.

- Sufficient conditions for uniform convergence can be found in Keener, Chapter 9, Theorem 9.2.
- For example, we have (a) if
 - Ω is compact,
 - $\theta \mapsto m_\theta(x)$ is continuous (for a.e x), and
 - $\mathbb{E}\|m_*(X_1)\|_\infty < \infty$.
$$(\|m_*(X_1)\|_\infty = \sup_{\theta \in \Omega} |m_\theta(X_1)|)$$
- For example, we have (b) if
 - Ω is compact, and
 - M is continuous, and
 - M has a unique maximizer over Ω .
- In general, the key factor in whether the uniform convergence holds is the size of the parameter space Ω .

Side note:

- Why do we have (b) if
 - Ω is compact, and
 - M is continuous, and
 - M has a unique maximizer over Ω ?
- Since Ω is compact and M is continuous, it attains its maximum over

$$\Omega \setminus B(\theta_0; \varepsilon) := \{\theta \in \Omega : d(\theta, \theta_0) \geq \varepsilon\}.$$

where $B(\theta_0; \varepsilon)$ is the open ball of radius ε centered at θ_0 .

- Let θ_ε be a maximizer of M over $\Omega \setminus B(\theta_0; \varepsilon)$. Then,

$$\sup_{\theta: d(\theta, \theta_0) \geq \varepsilon} M(\theta) = M(\theta_\varepsilon) < M(\theta_0)$$

- Strict inequality is due to the uniqueness of maximizer of M over Ω .
- Compactness is key, otherwise uniqueness of global maximizer does not imply this inequality.

Example 48

- MLE can be obtained as an M -estimator with $m_\theta(x) = \log \frac{p_\theta(x)}{p_{\theta_0}(x)}$.
- Addition of $-\log p_{\theta_0}(x)$ does not change the maximizer of $M_n(\theta)$.

$$M(\theta) = \mathbb{E}_{\theta_0}[m_\theta(X_1)] = - \int p_{\theta_0}(x) \log \frac{p_{\theta_0}(x)}{p_\theta(x)} dx = -D(p_{\theta_0} \| p_\theta).$$

- $D(p \| q)$ is the **Kullback–Leibler (KL) divergence** between p and q .
- A form of (squared) distance among distributions.
- Does not satisfy triangle equality or symmetry.
- $D(p \| q) \geq 0$ with equality iff $p = q$.
- Condition (b) is a bit stronger.
- Often, we can show (strong identifiability)

$$\gamma(d(\theta_0, \theta)) \leq D(p_{\theta_0} \| p_\theta)$$

for some strictly increasing function $\gamma \in [0, \infty) \rightarrow [0, \infty)$ in a neighborhood of θ_0 .

- Example: exponential distribution with $p_\lambda(x) = \lambda e^{-\lambda x} 1\{x > 0\}$:

$$\begin{aligned} D(p_{\lambda_0} \| p_\lambda) &= \mathbb{E}_{\lambda_0} \left[\log \frac{\lambda_0 e^{-\lambda_0 X_1}}{\lambda e^{-\lambda X_1}} \right] \\ &= \log \frac{\lambda_0}{\lambda} + \mathbb{E}_{\lambda_0} [(\lambda - \lambda_0) X_1] \\ &= -\log \frac{\lambda}{\lambda_0} + \frac{\lambda}{\lambda_0} - 1 \end{aligned}$$

- Itakura-Saito distance, or the Bregman divergence for $\phi(x) = -\log x$, from an earlier lecture.
- $f(x) = -\log x + x - 1$ is strictly convex on $(0, \infty)$ with unique minimum at $x = 1$.

First-order (asymptotic normality)

- More refined understanding, by looking at scaled (magnified) deviations of consistent estimators.
- IID sequence $X_1, X_n, \dots$ with mean $\mu = \mathbb{E}[X_1]$ and $\Sigma = \text{cov}(X_1)$:

WLLN $\bar{X}_n \xrightarrow{P} \mu$ $\bar{X}_n$ is consistent for μ .

CLT $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma)$ Characterizes fluctuations of $\bar{X}_n - \mu$.

- **Fluctuations are of the order $n^{-1/2}$** and after normalization have approximate Gaussian dist'n.

- First, let us look at how modes of convergence interact.

Proposition 12

- (a) $X_n \xrightarrow{p} X$ implies $X_n \xrightarrow{d} X$, but not vice versa.
- (b) $X_n \xrightarrow{p} c$ is equivalent to $X_n \xrightarrow{d} c$. (c is a constant.)
- (c) **Continuous mapping (CM)**: $X_n \rightarrow X$ and f is continuous, implies $f(X_n) \rightarrow f(X)$.
Holds for both $\xrightarrow{d}$ and $\xrightarrow{p}$.
- (d) **Slutsky's**: $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$ implies $(X_n, Y_n) \xrightarrow{d} (X, c)$.
- (e) $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$ implies $(X_n, Y_n) \xrightarrow{p} (X, Y)$.
- (f) $X_n \xrightarrow{d} X$ and $d(X_n, Y_n) \xrightarrow{p} 0$ implies $Y_n \xrightarrow{d} X$.

- For (c), f only need to be continuous on a set C with $\mathbb{P}(X \in C) = 1$.
- (d) does not hold in general if c is replaced by some random variable Y .

- Slutsky's lemma is usually not what we mentioned.
- It in fact is a special of application of (c) and (d), to functions $(x, y) \mapsto x + y$, $(x, y) \mapsto xy$ and $(x, y) \mapsto y^{-1}x$.

Corollary 5 (Slutsky's lemma)

Let X_n, Y_n and X be random variables, or vectors or matrices, and c a constant. Assume that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$. Then,

$$X_n + Y_n \xrightarrow{d} X + c, \quad Y_n X_n \xrightarrow{d} cX, \quad Y_n^{-1} X_n \xrightarrow{d} c^{-1}X,$$

assuming c is invertible for the latter. More generally $f(X_n, Y_n) \xrightarrow{d} f(X_n, c)$ for any continuous function.

E.g. $o_p(1) + o_p(1) = o_p(1)$.

- Simple examples:

(a) $X_n \xrightarrow{d} Z \sim N(0, 1)$ implies $X_n^2 \xrightarrow{d} Z^2 \sim \chi_1^2$.

Example 49 (Counterexample)

- $X_n = X \sim U(0, 1), \forall n$ and

$$Y_n = X_n 1\{n \text{ odd}\} + (1 - X_n) 1\{n \text{ even}\}.$$

- $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} X$, but (X_n, Y_n) does not converge in distribution.
- Why?
- Let $C_1 = \{(x, y) \in [0, 1]^2 : x = y\}$ and $C_2 = \{(x, y) \in [0, 1]^2 : x + y = 1\}$.
- Let $U(C_i)$ be uniform distribution on C_i . Then,

$$(X_n, Y_n) \sim \begin{cases} U(C_1) & n \text{ odd} \\ U(C_2) & n \text{ even} \end{cases}$$

Example 50 (t -statistic)

- IID sequence $\{X_i\}$, with $\mathbb{E}[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$.
- Let $\bar{X}_n = \frac{1}{n} \sum X_i$ and $S_n^2 = \frac{1}{n} \sum (X_i - \bar{X})^2 = \frac{1}{n} \sum X_i^2 - (\bar{X}_n)^2$. Then,

$$t_{n-1} := \frac{\bar{X}_n - \mu}{S_n / \sqrt{n}} \xrightarrow{d} N(0, 1).$$

- Why? $\frac{1}{n} \sum X_i^2 \xrightarrow{d} \mathbb{E}[X_1^2] = (\sigma^2 + \mu^2)$ and $\bar{X}_n^2 \xrightarrow{d} \mu^2$.
- These imply $S_n \xrightarrow{d} \sqrt{\sigma^2 + \mu^2 - \mu^2} = \sigma$.
- It follows that

$$t_{n-1} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{d} \frac{N(0, \sigma^2)}{\sigma} = N(0, 1)$$

- Distribution-free result: We are not assuming that $\{X_i\}$ are Gaussian.

- Also need the concept of **uniform tightness** or **boundedness in probability**.
- A collection of random vectors $\{X_n\}$ is uniformly tight if

$$\forall \varepsilon > 0, \exists M \text{ such that } \sup_n \mathbb{P}(\|X_n\| > M) < \varepsilon$$

We will write $X_n = O_p(1)$ in this case.

Proposition 13 (Uniform Tightness)

- (a) If $X_n \xrightarrow{P} 0$ and $\{Y_n\}$ is uniformly tight, then $X_n Y_n \xrightarrow{P} 0$.
- (b) If $X_n \xrightarrow{d} X$, then $\{X_n\}$ is uniformly tight.

(a) can be written compactly as $o_p(1)O_p(1) = o_p(1)$.

Simplified notation: $\mathbb{E}[\dot{m}_{\theta_0}]$ in place of $\mathbb{E}[\dot{m}_{\theta_0}(X_1)]$.

Theorem 13 (Asymptotic normality of M -estimators)

Assume the following

(a) $\dot{m}_{\theta_0}(X_1)$ has up to second moments with

- $\mathbb{E}[\dot{m}_{\theta_0}] = 0$, and
- well-defined covariance matrix $S_{\theta_0} := \mathbb{E}[\dot{m}_{\theta_0} \dot{m}_{\theta_0}^T]$.

(b) The Hessian $\ddot{m}_{\theta_0}(X_1)$ is integrable with $V_{\theta_0} := \mathbb{E}[\ddot{m}_{\theta_0}] \prec 0$.

(c) $\{\hat{\theta}_n\}$ is consistent for θ_0 .

(d) $\exists \varepsilon > 0$ such that $\sup_{\|\theta - \theta_0\| \leq \varepsilon} \|\ddot{M}_n(\theta) - \ddot{M}(\theta_0)\| \xrightarrow{P} 0$

Let $\Delta_{n,\theta} := \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{m}_{\theta}(X_i)$. Then,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1} \Delta_{n,\theta_0} + o_p(1), \quad \text{and} \quad \Delta_{n,\theta_0} \xrightarrow{d} N(0, S_{\theta_0}).$$

In particular, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V_{\theta_0}^{-1} S_{\theta_0} V_{\theta_0}^{-1})$.

In (b), only need the Hessian to be nonsingular.

(d) is (local) Uniform Convergence (UC)

Proof of AN

1. $\hat{\theta}_n$ is a maximizer of M_n , hence
2. $\dot{M}_n(\hat{\theta}_n) = 0$. (first-order optimality condition)
3. Taylor-expand $\dot{M}$ around θ_0 :

$$\dot{M}_n(\hat{\theta}_n) - \dot{M}_n(\theta_0) = \ddot{M}_n(\tilde{\theta}_n)[\hat{\theta}_n - \theta_0]$$

for some $\tilde{\theta}_n$ in the line segment $[\hat{\theta}_n, \theta_0]$.

(Mean-value theorem, assuming continuity of $\ddot{M}_n$)

4. $\tilde{\theta}_n = \theta_0 + o_p(1)$. (By consistency of $\hat{\theta}_n$)
5. $\ddot{M}_n(\tilde{\theta}_n) = \ddot{M}_n(\theta_0) + o_p(1)$ (By (d): UC)
6. Note that $\ddot{M}_n(\theta_0) = n^{-1} \sum_i \ddot{m}_{\theta_0}(X_i)$ is an average.
7. $\ddot{M}_n(\theta_0) = \mathbb{E}_{\theta_0}[\ddot{m}_{\theta_0}(X_1)] + o_p(1) = V_{\theta_0} + o_p(1)$. (By (b) and WLLN)
8. $\ddot{M}_n(\tilde{\theta}_n) = V_{\theta_0} + o_p(1)$. (Combine 5. and 7. + CM)
9. By CM applied w/ $f(X) = X^{-1}$, and invertibility of V_{θ_0} :

$$[\ddot{M}_n(\tilde{\theta}_n)]^{-1} = [V_{\theta_0} + o_p(1)]^{-1} = V_{\theta_0}^{-1} + o_p(1).$$

10. Combine 2., 3. and 9.

$$\begin{aligned}\hat{\theta}_n - \theta_0 &= [\ddot{M}_n(\tilde{\theta}_n)]^{-1} [\dot{M}_n(\hat{\theta}_n) - \dot{M}_n(\theta_0)] \\ &= [V_{\theta_0}^{-1} + o_p(1)] [0 - \dot{M}_n(\theta_0)]\end{aligned}$$

11. Expand RHS and multiply by $\sqrt{n}$,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1}[\sqrt{n}\dot{M}_n(\theta_0)] - o_p(1)[\sqrt{n}\dot{M}_n(\theta_0)] \quad (8)$$

12. $\dot{M}_n(\theta_0)$ is an average with zero-mean terms w/ covariance S_{θ_0} by (a).

13. $\sqrt{n}\dot{M}_n(\theta_0) \xrightarrow{d} N(0, S_{\theta_0})$. (CLT and (a))

14. $\sqrt{n}\dot{M}_n(\theta_0) = O_p(1)$ (By Prop. 13(b) and 13.)

15. Applying $o_p(1)O_p(1) = o_p(1)$ to (8), (Prop. 13(b) and 11.)

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1}[\sqrt{n}\dot{M}_n(\theta_0)] + o_p(1).$$

16. Note that $\sqrt{n}\dot{M}_n(\theta_0) = \Delta_{n,\theta}$ by definition.

17. Second part: Apply CM with $f(x) = -V_{\theta_0}^{-1}x$. (Exercise.)

Example 51 (AN of MLE)

- For MLE, $m_\theta(x) = \ell_\theta(x) = \log p_\theta(x)$.
- $\dot{m}_\theta = \dot{\ell}_\theta$, the score-function, zero-mean under regularity conditions.
- $S_\theta = \mathbb{E}_\theta[\dot{\ell}_\theta \dot{\ell}_\theta^T] = I(\theta)$.
- $V_\theta = \mathbb{E}_\theta[\ddot{\ell}_\theta] = -I(\theta)$
- Asymptotic covariance of MLE = $[-I(\theta)]^{-1} I(\theta) [-I(\theta)]^{-1} = [I(\theta)]^{-1}$.
- It follows (assuming (c) and (d) hold)

$$\sqrt{n}(\hat{\theta}_{\text{mle}} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$$

- Often interpreted as “MLE is asymptotically efficient”,
- i.e., achieves Cramer–Rao bound in the limit.

Hodge's superefficient example

- If $\sqrt{n}(\delta_n - \theta) \xrightarrow{d} N(0, \sigma^2(\theta))$, one might think that $\sigma^2(\theta) \geq 1/I(\theta)$ by CRB.
- If so, any estimator with variance $1/I(\theta)$ could be called asymptotically efficient.
- Unfortunately this is not true. (The convergence in distribution is rather weak to guarantee this.) Here is a counterexample:

Example 52

- Consider the shrinkage estimator

$$\delta'_n = \begin{cases} a\delta_n & |\delta_n| \leq n^{-1/4} \\ \delta_n & \text{otherwise} \end{cases}$$

- δ'_n has the same asymptotic behavior as δ_n for $\theta \neq 0$.
- Asymptotic behavior of δ'_n at $\theta = 0$ is the same as $a\delta_n$ which has asymptotic variance $a^2\sigma^2(\theta)$; this can be made arbitrarily small by choosing a sufficiently small.

- Delta method is a powerful extension of the CLT.
- Assume that $f : \Omega \rightarrow \mathbb{R}^k$, with $\Omega \subset \mathbb{R}^d$, is differentiable and $\theta \in \Omega$.
- Let $J_\theta = (\partial f_i / \partial x_j) |_{x=\theta}$ be the Jacobian of f at θ .
- Note: $J_\theta \in \mathbb{R}^{k \times d}$.

Proposition 14

Under the above assumption: If $a_n(X_n - \theta) \xrightarrow{d} Z$, with $a_n \rightarrow \infty$, then

$$a_n[f(X_n) - f(\theta)] \xrightarrow{d} J_\theta Z$$

- If f is differentiable, then it is partially differentiable and its total derivative can be represented (or identified) with the Jacobian matrix J_θ .
- Simplest case, $k = d = 1$: $a_n[f(X_n) - f(\theta)] \xrightarrow{d} f'(\theta)Z$

Proof of Delta method

- $a_n(X_n - \theta) = O_p(1)$ and since $a_n \rightarrow \infty$, we have $X_n - \theta = o_p(1)$.
- By differentiability (1st order Taylor expansion)

$$f(\theta + h) = f(\theta) + J_\theta h + R(h)\|h\|$$

where $R(h) = o(1)$. Define $R(0) = 0$ so that R is continuous at 0.

- Applying with $h = X_n - \theta$, we have

$$f(X_n) = f(\theta) + J_\theta(X_n - \theta) + R(X_n - \theta)\|X_n - \theta\|.$$

- Multiplying by a_n , we get

$$a_n[f(X_n) - f(\theta)] = J_\theta[a_n(X_n - \theta)] + R(X_n - \theta)\|a_n(X_n - \theta)\|$$

- $\|a_n(X_n - \theta)\| = O_p(1)$.

- Multiplying by a_n , we get

$$a_n[f(X_n) - f(\theta)] = J_\theta[a_n(X_n - \theta)] + \underbrace{R(X_n - \theta)}_{o_p(1)} \underbrace{\|a_n(X_n - \theta)\|}_{O_p(1)}$$

- $R(X_n - \theta) = o_p(1)$, and $J_\theta[a_n(X_n - \theta)] \xrightarrow{d} J_\theta Z$, both by CM.
- The result follows from $o_p(1)O_p(1) = o_p(1)$ and Prop 12(f).

Example 53

- Let X_i be iid with $\mu = \mathbb{E}[X_i]$ and $\sigma^2 = \text{var}[X_i]$.
- By CLT, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$.
- Consider the function $f(t) = t^2$. Then, by delta method

$$\sqrt{n}(f(\bar{X}_n) - f(\mu)) \xrightarrow{d} f'(\mu)N(0, \sigma^2),$$

that is,

$$\sqrt{n}[(\bar{X}_n)^2 - \mu^2] \xrightarrow{d} N(0, \sigma^2(2\mu)^2).$$

- For $\mu = 0$, we get the degenerate result that $\sqrt{n}(\bar{X}_n)^2 \xrightarrow{d} 0$.
- In this case, we need to scale the error further, i.e.,
- $n(\bar{X}_n)^2 \xrightarrow{d} \sigma^2\chi_1^2$, which follows from CLT $\sqrt{n}\bar{X} \xrightarrow{d} \sigma N(0, 1)$ and CM.

Example 54

- $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(p)$.
- By CLT $\sqrt{n}(\bar{X}_n - p) \xrightarrow{d} N(0, p(1 - p))$.
- Let $f(p) = p(1 - p)$.
- $f(\bar{X}_n)$ is a plugin estimator for the variance, and

$$\sqrt{n}(f(\bar{X}_n) - f(p)) \xrightarrow{d} N(0, (1 - 2p)^2 p(1 - p))$$

since $f'(x) = 1 - 2x$.

- Again at $p = 1/2$ this is degenerate and the convergence happens at a faster rate.

These examples can be dealt with using the following extension.

Proposition 15

Consider the scalar case $k = d = 1$. If $\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and f is twice differentiable with $f'(\theta) = 0$, then,

$$n[f(X_n) - f(\theta)] \xrightarrow{d} \frac{1}{2}f''(\theta)\sigma^2\chi_1^2$$

Informal Derivation:

- $f(X_n) - f(\theta) = \frac{1}{2}f''(\theta)(X_n - \theta)^2 + o((X_n - \theta)^2)$.
- Since $n(X_n - \theta)^2 \xrightarrow{d} (\sigma Z)^2$ where $Z \sim N(0, 1)$, we get the result.

Example 55 (Multivariate delta method)

- $S^2 = \frac{1}{n} \sum_i X_i^2 - (\bar{X}_n)^2$. Let

$$Z_n := \frac{1}{n} \sum_{i=1}^n \begin{pmatrix} X_i \\ X_i^2 \end{pmatrix}, \quad \theta = \begin{pmatrix} \mu \\ \mu^2 + \sigma^2 \end{pmatrix}, \quad \Sigma = \text{cov} \left(\begin{pmatrix} X_1 \\ X_1^2 \end{pmatrix} \right)$$

- By (multivariate) CLT, we have

$$\sqrt{n}(Z_n - \theta) \xrightarrow{d} N(0, \Sigma)$$

- Letting $f(x, y) = (x, y - x^2)$, we have

$$\sqrt{n} \left[\begin{pmatrix} \bar{X}_n \\ S_n^2 \end{pmatrix} - \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix} \right] \xrightarrow{d} J_\theta N(0, \Sigma) = N(0, J_\theta \Sigma J_\theta)$$

- Exercise: Evaluate asymptotic covariance $J_\theta \Sigma J_\theta$.

What are asymptotic normality results useful for?

- Simplify comparison of estimators: Can use asymptotic variances. (ARE)
- Can build asymptotic confidence intervals.

Asymptotic relative efficiency (ARE)

- Can compare estimators based on their asymptotic variance.
- Assume that for two estimators $\hat{\theta}_{1,n}$ and $\hat{\theta}_{2,n}$, we have

$$\sqrt{n}(\hat{\theta}_{i,n} - \mu(\theta)) \xrightarrow{d} N(0, \sigma_i^2(\theta)), \quad i = 1, 2.$$

- For large n , the variance of $\hat{\theta}_{i,n}$ is $\approx \sigma_i^2(\theta)/n$.
- Relative efficiency of $\hat{\theta}_{1,n}$ with respect to $\hat{\theta}_{2,n}$ can be measured in terms the ratio of the number of samples required to achieve the same asymptotic variance (i.e., error),

$$\frac{\sigma_1^2(\theta)}{n_1} = \frac{\sigma_2^2(\theta)}{n_2} \implies \text{ARE}_\theta(\hat{\theta}_1, \hat{\theta}_2) = \frac{n_2}{n_1} = \frac{\sigma_2^2(\theta)}{\sigma_1^2(\theta)}$$

If the above $\text{ARE} > 1$, then we prefer $\hat{\theta}_1$ over $\hat{\theta}_2$.

Example 56

- $X_i \stackrel{\text{iid}}{\sim} f_X$ with mean = (unique) median = θ , and variance 1.
- By CLT, we have $\sqrt{n}(\bar{X}_n - \theta) \rightarrow N(0, 1)$.
- Sample median: $Z_n = \text{median}(X_1, \dots, X_n) = X_{(\frac{1}{2}n)}$
- Can show

$$\sqrt{n}(Z_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{4[f_X(\theta)]^2}\right)$$

- Consider normal location family: $X_i \stackrel{\text{iid}}{\sim} N(\theta, 1)$.
- $f_X(\theta) = \phi(0) = 1/\sqrt{2\pi}$ where ϕ is the density of standard normal.
- Hence, $\sigma_{Z_n}^2(\theta) = \pi/2$.
- ARE of sample mean relative to median:

$$\frac{\sigma_{Z_n}^2(\theta)}{\sigma_{\bar{X}_n}^2(\theta)} = \frac{\pi}{2} \approx 1.57$$

- In normal family, we prefer the mean, since the median requires roughly 1.57 more samples to achieve the same accuracy.

Confidence intervals

An alternative to point estimators which provides a measure of our uncertainty or confidence. Recall $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{P}_{\theta_0}$.

Definition 17

A $(1 - \alpha)$ -confidence set for θ_0 is a random set $S = S(X_1, \dots, X_n)$ such that $\mathbb{P}_{\theta_0}(\theta_0 \in S) \geq 1 - \alpha$

- Trade-off between **size** of the set S and its **coverage probability** $\mathbb{P}_{\theta_0}(\theta_0 \in S)$.
- Want to minimize size while maintaining a lower bound on coverage prob.
- Usually CIs are built based on **pivots**:
- Functions of data and parameter whose dist'n is independent of param.

Example 57 (Normal family, known variance)

- $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then $Z = (\bar{X}_n - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$.
- Let $z_{\alpha/2}$ be such that $\mathbb{P}(Z \geq z_{\alpha/2}) = \alpha/2$.

$$\mathbb{P}(|\sqrt{n}(\bar{X}_n - \mu)| \leq z_{\alpha/2}) = 1 - \alpha \iff \mathbb{P}\left(\mu \in [\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}]\right) = 1 - \alpha.$$

Example 58 (Normal family, unknown variance)

- $X_i \sim N(\mu, \sigma^2)$.
- $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$
- $V = (n-1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$ where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.
- Hence, $T := Z/\sqrt{V/(n-1)} \sim t_{n-1}$ (Student's t distribution).
- Let t_α be such that $\mathbb{P}(|T| \geq t_{n-1}(\frac{\alpha}{2})) = \alpha$.
- $\bar{X}_n \pm t_{n-1}(\frac{\alpha}{2}) \frac{S_n}{\sqrt{n}}$ is an exact $(1 - \alpha)$ confidence interval.

Definition 18

An asymptotic $(1 - \alpha)$ -confidence set for θ_0 is a random set $S = S(X_1, \dots, X_n)$ such that $\mathbb{P}_{\theta_0}(\theta_0 \in S) \rightarrow 1 - \alpha$ as $n \rightarrow \infty$.

Example 59

- If $\sqrt{n}(T_n - \theta_0) \xrightarrow{d} N(0, \sigma^2(\theta_0))$, then assuming $\sigma(\cdot)$ is continuous.

$$T_n \pm \sqrt{\frac{\sigma^2(T_n)}{n}} z_{\alpha/2}$$

is an asymptotic C.I. at level $1 - \alpha$.

- Why? Since $T_n \xrightarrow{P} \theta_0$, by CM theorem, $\frac{\sigma(\theta_0)}{\sigma(T_n)} \xrightarrow{P} 1$. By Slutsky's lemma

$$\sqrt{\frac{n}{\sigma^2(T_n)}}(T_n - \theta_0) = \sqrt{\frac{\sigma^2(\theta_0)}{\sigma^2(T_n)}} \sqrt{\frac{n}{\sigma^2(\theta_0)}}(T_n - \theta_0) \xrightarrow{d} N(0, 1).$$

Example 60 (Asym. CI for MLE based on Fisher info)

- Recall that under regularity $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \frac{1}{I(\theta_0)})$, or

$$\sqrt{nl(\theta_0)}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 1).$$

- Assuming $I(\cdot)$ is continuous, applying the previous example,

$$\sqrt{nl(\hat{\theta}_n)}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 1).$$

- Hence, the following is asymptotic $(1 - \alpha)$ -CI for θ_0 :

$$\hat{\theta}_n \pm \frac{z_{\alpha/2}}{\sqrt{nl(\hat{\theta}_n)}}$$

Example 61 (Asym. CI based on empirical Fisher Info.)

- Let $\ell_n(\theta) = \sum_{i=1}^n \log p_\theta(X_i)$ and $I(\theta) = \mathbb{E}[-\frac{\partial^2}{\partial \theta^2}]$
- One can consider $-\frac{1}{n}\ddot{\ell}_n(\theta)$ as the empirical version of $I(\theta)$.
- (It is an unbiased and consistent estimate. $I(\theta)$ is Fisher info. based on sample of size 1.)
- By the same argument as in AN theorem, $-\frac{1}{n}\ddot{\ell}_n(\hat{\theta}_n) \xrightarrow{P} I(\theta_0)$.
- It follows that $\sqrt{\frac{-\ddot{\ell}_n(\hat{\theta}_n)}{nI(\theta_0)}} \xrightarrow{P} 1$
- By Slutsky's lemma,

$$\sqrt{\frac{-\ddot{\ell}_n(\hat{\theta}_n)}{nI(\theta_0)}} \sqrt{nI(\theta_0)}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 1)$$

- In other words, $\sqrt{-\ddot{\ell}_n(\hat{\theta}_n)}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 1)$.
- Hence, the following is asymptotic $(1 - \alpha)$ -CI for θ_0 :

$$\hat{\theta}_n \pm \frac{z_{\alpha/2}}{\sqrt{-\ddot{\ell}_n(\hat{\theta}_n)}}$$

Variance-stabilizing transform

- Assume $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2(\theta))$.
- By delta method, $\sqrt{n}[f(T_n) - f(\theta)] \xrightarrow{d} N(0, [f'(\theta)]^2 \sigma^2(\theta))$.
- We can choose f so that $[f'(\theta)]^2 \sigma^2(\theta) = C$, a constant.
- Good for building asymptotic pivots.

Example 62

- $X_i \stackrel{\text{iid}}{\sim} \text{Poi}(\theta)$. Note $\mathbb{E}_\theta[X_i] = \text{var}_\theta[X_i] = \theta$.
- By CLT $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \theta)$.
- Take $f'(\theta) = \frac{1}{\sqrt{\theta}}$. Can be realized by $f(\theta) = 2\sqrt{\theta}$, hence

$$2\sqrt{n}(\sqrt{\bar{X}_n} - \sqrt{\theta}) \xrightarrow{d} N(0, 1)$$

- Asymptotic CI for $\sqrt{\theta}$ of level $1 - \alpha$: $(\sqrt{\bar{X}_n} \pm \frac{1}{2\sqrt{n}} z_{\alpha/2})$.
- Compare with standard asym. CI for θ : $(\bar{X}_n \pm \sqrt{\frac{\bar{X}_n}{n}} z_{\alpha/2})$.

Hypothesis testing

- Recall decision theory framework:
- Probabilistic model $\{\mathbb{P}_\theta : \theta \in \Omega\}$ for $X \in \mathcal{X}$.
- Special case : Ω is partitioned into two disjoint sets Ω_0 and Ω_1 .
- Want to decide which piece θ belongs.
- Can form an estimate $\hat{\theta}$ for θ and output $1\{\hat{\theta} \in \Omega_1\}$.
- A general principal:
 - Do not estimate more than what you care about.*
 - The more complex the model, the more potential for fitting to noise.*

- We want to test

$$\begin{aligned}H_0 &: \theta \in \Omega_0 \quad (\text{null}) \\H_1 &: \theta \in \Omega_1 \quad (\text{alternative})\end{aligned}$$

- A non-random test can be specified with a critical region $S \subset \mathcal{X}$ as

$$\delta(X) = 1\{X \in S\}.$$

- When $\delta(X) = 1$, we have accepted H_1 , or “rejected H_0 ”.
- Power function of the test is given by

$$\beta(\theta) = \mathbb{P}_\theta(X \in S) = \mathbb{E}_\theta[\delta(X)]$$

- We would like $\beta(\theta) \approx 1\{\theta \in \Omega_1\}$.
- It cannot be achieved, so we settle for a trade-off. Define

$$\text{significance level} \quad \alpha = \sup_{\theta \in \Omega_0} \beta(\theta)$$

$$\text{power of the test} \quad \beta = \inf_{\theta \in \Omega_1} \beta(\theta)$$

- Neyman–Pearson framework: Maximize β subject to a fixed α .

- Often need to consider a **randomized test**, in which case interpret

$$\delta(x) = \mathbb{P}(\text{accept } H_1 \mid X = x)$$

- Power function $\beta(\theta) = \mathbb{E}_\theta[\delta(X)]$ still gives the probability of accepting H_1 , by the smoothing property.

Simple hypothesis test

- $\Omega_0 = \{\theta_0\}$ and $\Omega_1 = \{\theta_1\}$.
- Neyman–Pearson criterion reads: Fix α

$$\sup_{\delta} \mathbb{E}_{\theta_1}[\delta(X)] \quad \text{s.t.} \quad \mathbb{E}_{\theta_0}[\delta(X)] \leq \alpha.$$

Most powerful (MP) test for significance level at most α .

- Neyman–Pearson lemma:
Most power achieved by a likelihood ratio test (LRT),

$$\delta(X) = 1\{L(x) > \tau\} + \gamma 1\{L(x) = \tau\}, \quad L(x) := p_{\theta_1}(x)/p_{\theta_0}(x).$$

- Sometimes write $1\{p_{\theta_1}(x) \geq \tau p_{\theta_0}(x)\}$ to avoid division by zero.
- For simplicity assume write $p_0 = p_{\theta_0}$ and $p_1 = p_{\theta_1}$.
- So we write $L(x) = p_1(x)/p_0(x)$ for example.

Informal proof

- For simplicity drop dependence on X : $\delta = \delta(X)$ and $L = L(X)$.
- Introduce **Lagrange multipliers**, and solve the unconstrained problem:

$$\delta^* = \operatorname{argmax}_{\delta} \left[\mathbb{E}_1(\delta) + \lambda(\alpha - \mathbb{E}_0(\delta)) \right] = \operatorname{argmax}_{\delta} \left[\mathbb{E}_1(\delta) - \lambda \mathbb{E}_0(\delta) \right]$$

- Recall the **change of measure** formula (note $L = p_1/p_0$):

$$\mathbb{E}_1[\delta] = \int \delta p_1 d\mu = \int \delta L p_0 d\mu = \mathbb{E}_0[\delta L].$$

- The problem reduces to

$$\delta^* = \operatorname{argmax}_{\delta} \mathbb{E}_0[\delta(L - \lambda)]$$

- The optimal solution is

$$\delta^* = \begin{cases} 1 & L > \lambda \\ 0 & L < \lambda \end{cases}$$

which is a likelihood ratio test.

Theorem 14 (Nyeman-Pearson Lemma)

Consider the family of (randomized) likelihood ratio tests

$$\delta_{t,\gamma}(x) = \begin{cases} 1 & p_1(x) > t p_0(x) \\ \gamma & p_1(x) = t p_0(x) \\ 0 & p_1(x) < t p_0(x) \end{cases}$$

The following hold:

- (a) *For every $\alpha \in [0, 1]$, there are t, γ such that $\mathbb{E}_0[\delta_{t,\gamma}(X)] = \alpha$.*
- (b) *If a LRT satisfies $\mathbb{E}_0[\delta_{t,\gamma}(X)] = \alpha$, then it is most powerful (MP) at level α .*
- (c) *Any MP test at level α can be written as a LRT.*

- Part (a) follows by looking at $g(t) = \mathbb{P}_0(L(X) > t) = 1 - F_Z(t)$ where $Z = L(X)$. g is non-increasing and right-continuous, etc. (Draw a picture.)

Proof of Neyman-Pearson Lemma

- For part (b), let δ^* be the LRT with significance level α .
- Let δ be any other rule satisfying $\mathbb{E}_0[\delta(X)] \leq \alpha = \mathbb{E}_0[\delta^*(X)]$.
- For all x , (consider the three possibilities)

$$\delta(x)[p_1(x) - tp_0(x)] \leq \delta^*(x)[p_1(x) - tp_0(x)]$$

- Integrate w.r.t. x :

$$\mathbb{E}_1[\delta(X)] - t\mathbb{E}_0[\delta(X)] \leq \mathbb{E}_1[\delta^*(X)] - t\mathbb{E}_0[\delta^*(X)]$$

or

$$\mathbb{E}_1[\delta(X)] - \mathbb{E}_1[\delta^*(X)] \leq t(\mathbb{E}_0[\delta(X)] - \mathbb{E}_0[\delta^*(X)]) \leq 0$$

- Conclude that $\mathbb{E}_1[\delta(X)] \leq \mathbb{E}_1[\delta^*(X)]$.
- Part (c), left as an exercise.

Example 63

- Consider $X \sim N(\theta, 1)$ and the two hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1$$

- Likelihood ratio is

$$L(x) = \frac{p_1(x)}{p_0(x)} = \frac{\exp[-\frac{1}{2}(x - \theta_1)^2]}{\exp[-\frac{1}{2}(x - \theta_0)^2]}$$

- LRT rejects H_0 if $L(x) > t$. Equivalently

$$\begin{aligned} \log L(x) > \log t &\iff -\frac{1}{2}(x - \theta_1)^2 + \frac{1}{2}(x - \theta_0)^2 > \log t \\ &\iff x(\theta_1 - \theta_0) + \frac{1}{2}(\theta_0^2 - \theta_1^2) > \log t \\ &\iff x \cdot \text{sign}(\theta_1 - \theta_0) > \frac{\log t - \frac{1}{2}(\theta_0^2 - \theta_1^2)}{|\theta_1 - \theta_0|} =: \tau \end{aligned}$$

- Assume $\theta_1 > \theta_0$. Then, the test is equivalent to $x > \tau$.
- We set τ by requiring $\mathbb{P}_0(X > \tau) = \alpha$. This gives $\tau = \theta_0 + Q^{-1}(\alpha)$.

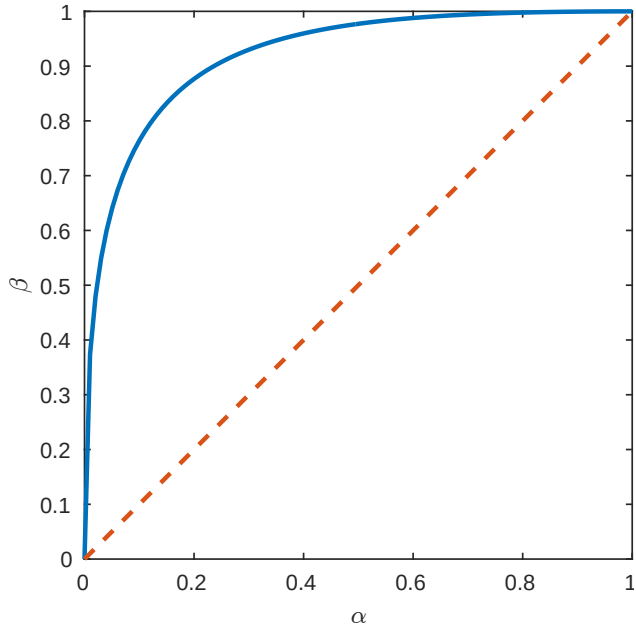
Power calculation (Previous example continued)

- $Q(x) = 1 - \Phi(x)$ where Φ is the CDF of standard normal distribution.
- The power is (since $X - \theta_1 \sim N(0, 1)$ under $\mathbb{P}_1$)

$$\beta = \mathbb{P}_1(X > \tau) = \mathbb{P}_1(X - \theta_1 > \tau - \theta_1) = Q(\tau - \theta_1)$$

- Plugging in τ , we have $\beta = Q(-\delta + Q^{-1}(\alpha))$ where $\delta = \theta_1 - \theta_0$.
- Plot of β versus α is the **ROC curve** of the test.
- ROC = Receiver Operating Characteristic
- See next slide.
- Alternatively, can plot parametric curve $\beta = Q(\tau - \theta_1)$ and $\alpha = Q(\tau - \theta_1)$, where parameter τ varies in $\mathbb{R}$.
- ROC of no test can go about this curve (by Neyman-Pearson lemma).

ROC curve



Composite hypothesis testing

- Often want to test $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \in \Omega_1$ where $\Omega_0 \cap \Omega_1 = \emptyset$.

Example 64

Testing whether a coin is fair or not. Here $\Omega_0 = \{1/2\}$ and $\Omega_1 = [0, 1] \setminus \{1/2\}$.

Definition 19

A test δ of size α is uniformly most powerful (UMP) at level α if

$$\forall \text{ tests } \phi \text{ of level } \leq \alpha, \forall \theta \in \Omega_1, \quad \beta_\delta(\theta) \geq \beta_\phi(\theta).$$

- UMP tests do not always exist (they often don't in fact).

Example 65 (Coin flipping continued)

- Observe $X \sim \text{Bin}(n, \theta)$.
- Consider testing $H_0 : \theta = 1/2$ versus $H_1 : \theta = \theta_1$ based on X .
- Most powerful test is a LRT, based on

$$T(x) = \frac{\theta_1^x (1 - \theta_1)^{n-x}}{(1/2)^x (1/2)^{n-x}} = \left(\frac{\theta_1}{1 - \theta_1} \right)^x \left(\frac{1 - \theta_1}{1/2} \right)^n$$

- Nature of the test changes based on whether $\theta_1 < 1/2$ or $\theta_1 > 1/2$:

$$\theta_1 < 1/2 \implies \log[\theta_1/(1 - \theta_1)] < 0 \implies \text{reject } H_0 \text{ when } x < \tau$$

$$\theta_1 > 1/2 \implies \log[\theta_1/(1 - \theta_1)] > 0 \implies \text{reject } H_0 \text{ when } x > \tau$$

- Suggests that a special structure is needed for the existence of a UMP test.

Definition 20

A family $\mathcal{P} = \{p_\theta(x) : \theta \in \Omega\}$ of densities has a monotone likelihood ratio (MLR) in $T(x)$ if for $\theta_0 < \theta_1$, the LR $L(x) = p_{\theta_1}(x)/p_{\theta_0}(x)$ is a non-decreasing function of $T(x)$.

For example, in the coin flipping problem, the model has MLR in $T(X) = X$.

Example 66 (1-D exponential family)

- Consider $p_\theta(x) = h(x) \exp[\eta(\theta)T(x) - A(\theta)]$. LR is

$$L(x) = \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)} = \exp [(\eta(\theta_1) - \eta(\theta_0))T(x) - A(\theta_1) + A(\theta_0)].$$

- If η is monotone (e.g. $\theta_0 \leq \theta_1 \implies \eta(\theta_0) \leq \eta(\theta_1)$), then the family has MLR in $T(x)$ or $-T(x)$.
- Includes the Bernoulli (or binomial) example before with $\eta(\theta) = \log \frac{\theta}{1-\theta}$. Others cases: normal location family, Poisson and exponential.

Example 67 (Non-exponential family)

- $X_i \stackrel{\text{iid}}{\sim} U[0, \theta]$, $i = 1, \dots, n$.
- $p(x) = \theta^{-n} 1\{x_{(n)} \leq \theta\} 1\{x_{(1)} \geq 0\}$, and

$$L(x) = \left(\frac{\theta_1}{\theta_0}\right)^n \frac{1\{x_{(n)} \leq \theta_1\}}{1\{x_{(n)} \leq \theta_0\}} = \begin{cases} \left(\frac{\theta_1}{\theta_0}\right)^n, & x_{(n)} \in [0, \theta_0) \\ \infty & x_{(n)} \in [\theta_0, \theta_1) \end{cases}$$

- Consider $\theta_1 > \theta_0$.
- For $x_{(n)} \in (0, \theta_1)$, $L(x)$ increasing in $x_{(n)}$ (transitions from $(\theta_1/\theta_0)^n$ to ∞).

Theorem 15 (UMP for one-sided problems)

- Let $\mathcal{P}$ be a family with MLR in $T(x)$.
- Consider the one-sided test $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.
- Then, $\delta(x) = 1\{T(x) > C\} + \gamma 1\{T(x) = C\}$, for γ, C such that $\beta_\delta(\theta_0) = \alpha$ is UMP of size α .

- Take $\theta_1 > \theta_0$ and let $L_{\theta_1, \theta_0}(x) = p_{\theta_1}(x)/p_{\theta_0}(x)$ be the corresponding LR.
- By the MLR property, the given test is a LR test, i.e.,

$$\delta(x) = 1\{L_{\theta_1, \theta_0}(x) > C'\} + \gamma 1\{L_{\theta_1, \theta_0}(x) = C'\}$$

for some constant $C' = C'(\theta_1, \theta_0)$.

- Since $\beta_\delta(\theta_0) = \alpha$, by Neyman–Pearson lemma, δ is MP for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$
- Since $\theta_1 > \theta_0$ was arbitrary, δ is UMP for $\theta = \theta_0$ versus $\theta > \theta_0$.
- Last piece to check is $\beta_\delta(\theta) \leq \alpha$ for $\theta < \theta_0$. (Exercise.)

Example 68 (Non-exponential family (continued))

- $X_i \stackrel{\text{iid}}{\sim} U[0, \theta]$, $i = 1, \dots, n$.
- The family has MLR in $X_{(n)}$.
- $\delta(X) = 1\{X_{(n)} \geq t\}$ is UMP for one-sided testing: $\theta > \theta_0$ against $\theta \leq \theta_0$.

- To set the threshold

$$g(t) = 1 - \mathbb{P}_{\theta_0}(X_{(n)} \leq t) = 1 - \prod_i \mathbb{P}_{\theta_0}(X_i \leq t) = \begin{cases} 1 - (t/\theta_0)^n, & t \leq \theta_0 \\ 0 & t > \theta_0 \end{cases}$$

which is a continuous function.

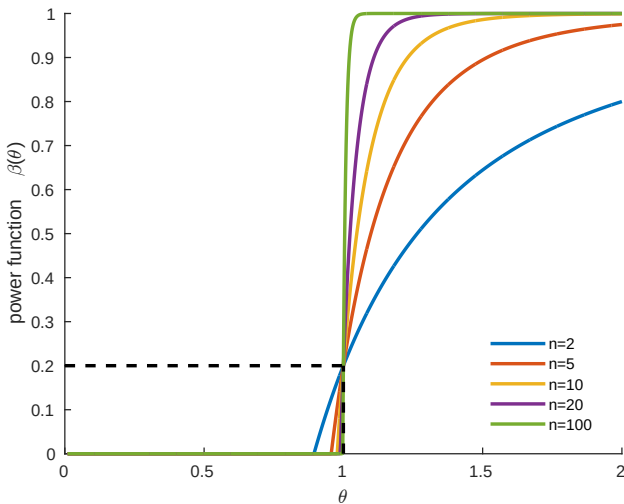
- Solving $g(t) = \alpha$ gives $t = (1 - \alpha)^{1/n} \theta_0$.
- Similarly the power function is

$$\beta(\theta) = \mathbb{P}_{\theta}(X_{(n)} > t) = [1 - (t/\theta)^n]_+$$

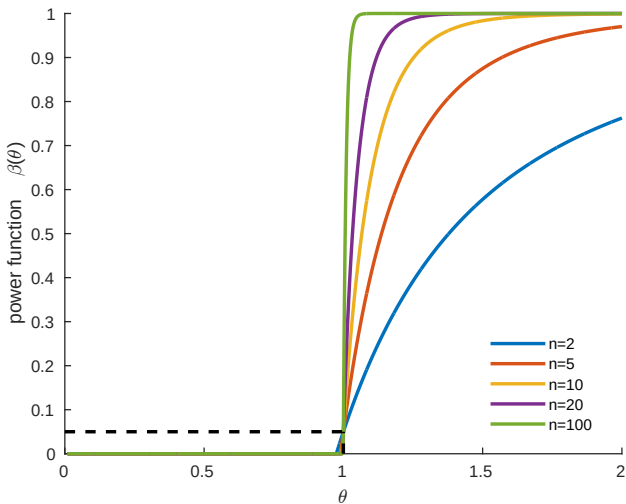
which holds for all $\theta > 0$.

- For the UMP test, we have $\beta(\theta) = [1 - (1 - \alpha)(\theta_0/\theta)^n]_+$.

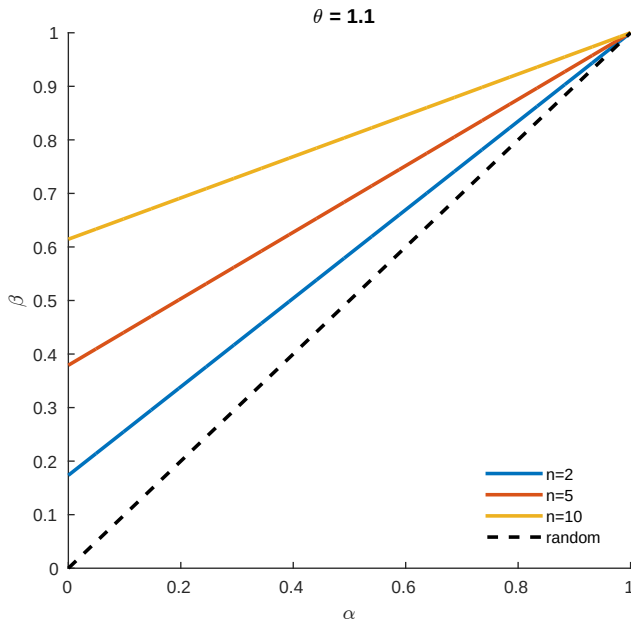
- Plots of the power function for $U[0, \theta]$ example for various n , ($\theta_0 = 1, \alpha = 0.2$).



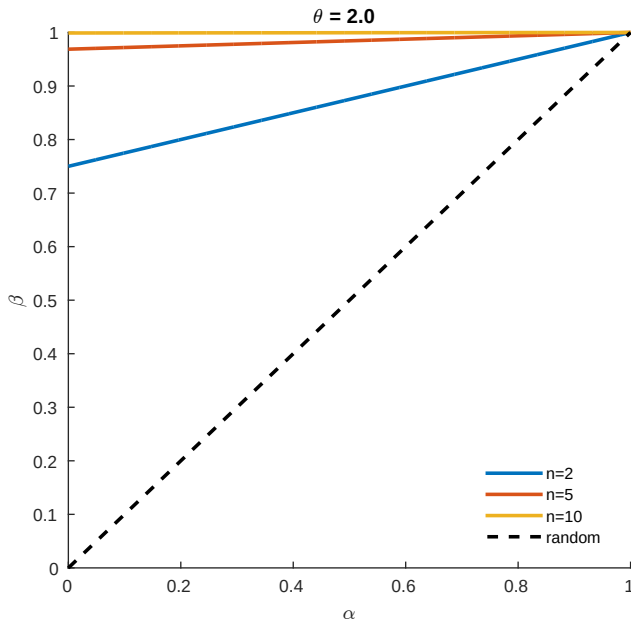
- Plots of the power function for $U[0, \theta]$ example for various n , ($\theta_0 = 1, \alpha = 0.05$).



- ROC plots for $U[0, \theta]$ example for various n , ($\theta_0 = 1, \theta = 1.1$).



- ROC plots for $U[0, \theta]$ example for various n , ($\theta_0 = 1, \theta = 2$).



Generalized likelihood ratio test (GLRT)

- Consider testing $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \in \Omega_1$.
- In the absence of UMPs a natural extension of LRT is the following generalized LRT

$$L(x) = \frac{\sup_{\theta \in \Omega} p_{\theta}(x)}{\sup_{\theta \in \Omega_0} p_{\theta}(x)} = \frac{p_{\hat{\theta}}(x)}{p_{\hat{\theta}_0}(x)} \in [1, \infty]$$

where $\Omega = \Omega_0 \cup \Omega_1$.

- $\hat{\theta}$ is the **unconstrained MLE**, whereas $\hat{\theta}_0$ is the **constrained MLE**, the maximizer of $\theta \mapsto p_{\theta}(x)$ over $\theta \in \Omega_1$.
- A GLRT rejects H_0 if $L(x) > \lambda$ for some threshold.
- Alternatively, GLRT can be written as

$$\delta(x) = 1\{\Lambda_n(x) \geq \tau\} + \gamma 1\{\Lambda_n(x) = \tau\}, \quad \Lambda_n(x) = 2 \log L(x)$$

- The threshold τ is set as usual by solving $\sup_{\theta \in \Omega_0} \mathbb{E}_{\theta}[\delta(X)] = \alpha$

- Why the above is a reasonable test?
- Assume $\gamma = 0$: No randomization is needed.
- When $\theta \in \Omega_0$, then both $\hat{\theta}$ and $\hat{\theta}_0$ approach θ as $n \rightarrow \infty$.
- Hence, $L(x) \approx 1$ when $\theta \in \Omega_0$.
- However, when $\theta \in \Omega_1$, the unconstrained MLE $\hat{\theta}$ approaches θ as $n \rightarrow \infty$, while $\hat{\theta}_0$ does not. This is because $\hat{\theta}_0 \in \Omega_0$ and $\theta \in \Omega_1$, and Ω_0 and Ω_1 are disjoint.
- It follows that $L(x) > 1$ when $\theta \in \Omega_1$ (in fact $L(x) \gg 1$ usually).
- By thresholding $L(x)$ at some $\lambda > 1$, we can tell the two hypotheses apart

Example 69

- $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, both μ and σ^2 unknown. Let $\theta = (\mu, \sigma^2)$. Take

$$\Omega = \{(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 > 0\}, \quad \Omega_0 = \{\theta_0\}, \text{ for } \theta_0 = (\mu_0, \sigma_0^2) = (0, 1).$$

- Want to test Ω_0 against $\Omega_1 = \Omega \setminus \Omega_0$.

$$\sup_{\theta \in \Omega_0} p_{\theta}(x) = p_{\theta_0}(x) = \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{1}{2} \sum_i x_i^2\right)$$

$$\sup_{\theta \in \Omega} p_{\theta}(x) = p_{\hat{\theta}}(x) = \frac{1}{(2\pi\hat{\sigma}^2)^{n/2}} \exp\left(-\frac{1}{2\hat{\sigma}^2} \sum_i (x_i - \hat{\mu})^2\right)$$

where $\hat{\theta} = (\hat{\mu}, \hat{\sigma}^2)$ with $\hat{\mu} = \frac{1}{n} \sum_i x_i$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \hat{\mu})^2$ is the MLE.

Example continued

- The GLR is

$$L(x) = \frac{p_{\hat{\theta}}(x)}{p_{\theta_0}(x)} = (\hat{\sigma}^2)^{-n/2} \exp \left[\frac{1}{2} \sum_i x_i^2 - \underbrace{\frac{1}{2\hat{\sigma}^2} (x_i - \hat{\mu})^2}_{n/2} \right].$$

The GLRT rejects H_0 when $L(x) > t_\alpha$ where $\mathbb{P}_{\theta_0}(L(x) > t_\alpha) = \alpha$.

- Alternatively, threshold

$$\Lambda_n(x) = 2 \log L(x) = -n \log \hat{\sigma}^2 + \sum_i x_i^2 - n.$$

- Will see that Λ_n in general has χ_r^2 distribution where r is the difference between the dimensions of the full (Ω) versus null (Ω_0) parameter spaces.
- Thus, we expect Λ_n in this problem to have an asymptotic χ_2^2 distribution under the null $\theta = \theta_0$. (It is instructive to try to show this directly.)

Asymptotics of GLRT

- Consider $\Omega \subset \mathbb{R}^d$, open, and let $r \leq d$. Take the null hypothesis to be

$$\begin{aligned}\Omega_0 &= \{\theta \in \Omega : \theta_1 = \theta_2 = \dots = \theta_r = 0\} \\ &= \{\theta \in \Omega : \theta = (0, \dots, 0, \theta_{r+1}, \dots, \theta_{r+d-r})\}.\end{aligned}$$

- Note that Ω_0 is a $(d - r)$ -dimensional subspace of Ω .

Theorem 16

Under the same assumptions guaranteeing asymptotic normality of MLEs,

$$\Lambda_n = 2 \log L(X) \xrightarrow{d} \chi_r^2, \quad \text{under } H_0.$$

*Degrees of freedom $r = d - (d - r)$, that is, the difference in the **local dimensions of full and null parameter sets**.*

- Recall $\ell_\theta(X) = \log p_\theta(X)$ and let $M_n(\theta) = \frac{1}{n} \sum_i \ell_\theta(X_i)$.
- We have $\Lambda_n = -2n[M_n(\hat{\theta}_{0,n}) - M_n(\hat{\theta}_n)]$.
- By Taylor expansion around $\hat{\theta}_n$ (unrestricted MLE), for some $\tilde{\theta}_n \in [\hat{\theta}_n, \hat{\theta}_{0,n}]$,

$$M_n(\hat{\theta}_{0,n}) - M_n(\hat{\theta}_n) = [\dot{M}_n(\hat{\theta}_n)]^T (\hat{\theta}_{0,n} - \hat{\theta}_n) + \frac{1}{2} (\hat{\theta}_{0,n} - \hat{\theta}_n)^T \ddot{M}_n(\tilde{\theta}_n) (\hat{\theta}_{0,n} - \hat{\theta}_n)$$

- Since $\hat{\theta}_n$ is the MLE, we have $\dot{M}_n(\hat{\theta}_n) = 0$ assuming $\hat{\theta}_n \in \text{int}(\Omega)$.
- By the same uniform arguments $\tilde{\theta}_n = \theta_0 + o_p(1)$ implies
- $\ddot{M}_n(\tilde{\theta}_n) = \ddot{M}_n(\theta_0) + o_p(1) = -I_{\theta_0} + o_p(1)$,
- the last equality is because $\ddot{M}_n(\theta_0) \xrightarrow{P} \mathbb{E}_{\theta_0}[\ddot{\ell}_{\theta_0}] = -I_{\theta_0}$ by WLLN.

- Assuming that $\sqrt{n}(\hat{\theta}_{0,n} - \hat{\theta}_n) = O_p(1)$, we obtain

$$\Lambda_n = [\sqrt{n}(\hat{\theta}_{0,n} - \hat{\theta}_n)]^T I_{\theta_0} \sqrt{n}(\hat{\theta}_{0,n} - \hat{\theta}_n) + o_p(1).$$

- Asymptotically GLR measures a particular distance (squared) between $\hat{\theta}_{0,n}$ and $\hat{\theta}_n$, one which weighs different directions differently, according to eigenvectors of $I(\theta_0)$.
- More specifically, let $\|z\|_Q := \sqrt{z^T Q z} = \|Q^{1/2} z\|_2^2$, which defines a norm when $Q \succ 0$. Then,

$$\Lambda_n = \|\sqrt{n}(\hat{\theta}_{0,n} - \hat{\theta}_n)\|_{I_{\theta_0}}^2 + o_p(1).$$

- In the simple case where $\Omega_0 = \{\theta_0\}$, we have $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I_{\theta_0}^{-1})$.
- Equivalently, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} I_{\theta_0}^{-1/2} Z$ where $Z \sim N(0, I_d)$.
- It follows from the CM theorem (since $z \mapsto \|z\|_Q$ is continuous) that

$$\Lambda_n \xrightarrow{d} \|I_{\theta_0}^{-1/2} Z\|_{I_{\theta_0}}^2 = \|I_{\theta_0}^{1/2} I_{\theta_0}^{-1/2} Z\|_2^2 = \|Z\|_2^2.$$

- Since $\|Z\|_2^2 = \sum_{i=1}^d Z_i^2 \sim \chi_d^2$ we have the proof $\Omega_0 = \{\theta_0\}$.
- The proof of the general case is more complicated and is omitted.

Example 70 (Multinomial: testing uniformity)

- $(X_1, \dots, X_d) \sim \text{Multinomial}(n, \theta)$, where $\theta = (\theta_1, \dots, \theta_d)$ is a probability vector, that is,

$$\theta \in \Omega = \{\theta \in \mathbb{R} : \theta_i \geq 0, \sum_i \theta_i = 1\}$$

- X_j counts how many of n objects fall into category j .

$$p_{\theta}(x) = \binom{n}{x_1, \dots, x_d} \prod_{i=1}^d \theta_i^{x_i} \propto \prod_{i=1}^d \theta_i^{x_i}.$$

- Would like to test $\Omega_0 = \{\theta_0\}$ where $\theta_0 = (\frac{1}{d}, \dots, \frac{1}{d})$ versus $\Omega_1 = \Omega \setminus \Omega_0$.
- MLE over Ω is given by $\hat{\theta}_i = x_i/n$. Requires techniques for constrained optimization, such as Lagrange multipliers, since Ω itself is constrained. (Exercise.)

- MLE over Ω is given by $\hat{\theta}_i = x_i/n$. This requires using techniques for constrained optimization, such as Lagrange multipliers, since Ω itself is constrained. (Exercise.)
- We obtain

$$\begin{aligned}\Lambda_n &= 2 \log \frac{p_{\hat{\theta}}(x)}{p_{\theta_0}(x)} = 2 \log \prod_{i=1}^d \left(\frac{\hat{\theta}_i}{(\theta_0)_i} \right)^{x_i} = 2 \sum_{i=1}^d x_i \log \frac{\hat{\theta}_i}{(\theta_0)_i} \\ &= 2n \sum_i \hat{\theta}_i \log \frac{\hat{\theta}_i}{(\theta_0)_i} = 2n D(\hat{\theta} \| \theta_0)\end{aligned}$$

- Both $\hat{\theta}$ and θ_0 are probability vectors; $D(\hat{\theta} \| \theta_0)$ is their KL divergence.
- GLRT does a sensible test: Reject null if $\hat{\theta}$ is far from θ_0 in KL divergence.
- Our asymptotic theory implies: $\Lambda_n \xrightarrow{d} \chi_{d-1}^2$ under the null, i.e. $\theta = \theta_0$, since Ω is $(d-1)$ -dimensional and Ω_0 is 0-dimensional. This is a fairly non-trivial result.

- Consider a family of tests $\delta_\alpha, \alpha \in (0, 1)$ indexed by their level α .
- Assume: $\alpha \mapsto \delta_\alpha(x)$ is nondecreasing, and right-continuous.
- E.g., if $\delta_\alpha(x) = 1\{x \in C(\alpha)\}$, then $C(\alpha_1) \subseteq C(\alpha_2)$ if $\alpha_1 \leq \alpha_2$.
- p -value or attained significance for observed x is defined as

$$p(x) := \inf \{ \alpha : \delta_\alpha(x) = 1 \}$$

- Note that $p = p(X)$ is a random variable. We have

$$p(X) \leq \alpha \iff \delta_\alpha(X) = 1$$

$p \leq \alpha$ implies $1 = \delta_p \leq \delta_\alpha$, since the infimum is attained by assumptions on δ_α , hence $\delta_\alpha = 1$. The other direction follows from the definition of inf.

- This implies $\mathbb{P}_0(p(X) \leq \alpha) = \mathbb{P}_0(\delta_\alpha(X) = 1) = \alpha$.
- That is, $p = p(X) \sim U[0, 1]$ under null.

Example 71 (Normal example continued)

- Consider $X \sim N(\theta, 1)$ and $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.
- MP test at level α is $\delta_\alpha(X) = 1\{X - \theta_0 \geq Q^{-1}(\alpha)\}$.
- Alternatively $\delta_\alpha(X) = 1\{Q(X - \theta_0) \leq \alpha\}$ since Q is decreasing.
- The p -value is

$$p(X) = \inf\{\alpha : Q(X - \theta_0) \leq \alpha\} = Q(X - \theta_0) \quad (9)$$

- Under null $X - \theta_0 \sim N(0, 1)$ hence $\Phi(X - \theta_0) \sim U[0, 1]$ (why?).
- Then, $p(X) = 1 - \Phi(X - \theta_0) \sim U[0, 1]$ as expected.

- Exercise: Verify that under $H_1 : \theta = \theta_1$, the CDF of $p(X)$ is

$$\mathbb{P}_1(p(X) \leq t) = Q(-\delta + Q^{-1}(t)).$$

where $\delta = \theta_1 - \theta_0$. Note that this curve is **the same as the ROC**.

- Recall $Q(t) = 1 - \Phi(t)$ where Φ is the CDF of $N(0, 1)$.

- Can verify that definition (9) produces the “common” definition of p -values, say when $\delta_\alpha(X) = 1\{T \geq \tau_\alpha\}$ or $\delta_\alpha(X) = 1\{|T| \geq \tau_\alpha\}$.

Example 72

- Consider the two-sided test and let $G(t) = \mathbb{P}_0(|T| \geq t)$.
- Assume that G is continuous hence invertible (both decreasing functions.)
- Requiring level α : $G(\tau_\alpha) = \mathbb{P}_0(|T| \geq \tau_\alpha) = \alpha \implies \tau_\alpha = G^{-1}(\alpha)$.
- This gives $\delta_\alpha(X) = 1\{|T| \geq G^{-1}(\alpha)\} = 1\{G(|T|) \leq \alpha\}$.
- By definition (9), $p = G(|T|)$ which is the common definition.

Multiple Hypothesis testing

- We have a collection of null hypotheses $H_{0,i}, i = 1, \dots, n$.

Example 73 (Basic example)

- Testing in the *normal means model*

$$y_i \sim N(\mu_i, 1), \quad i = 1, \dots, n$$

and $H_{0,i} : \mu_i = 0$.

- y_i could be the expression level (or state) of gene i .
 - $H_{0,i}$ means that gene i has no effect on the disease under consideration.
-
- Suppose that for each $H_{0,i}$, we have a test, hence a *p-value* p_i .
 - Assume under $H_{0,i} : p_i \sim U[0, 1]$.
 - A test that reject $H_{0,i}$ when $p_i \leq \alpha$, is of size α under i th null.

Testing global null

- The global null is $H_0 = \bigcap_{i=1}^n H_{0,i}$.
- Want to combine $p_1, \dots, p_n$ to build a test of level α for H_0 .
- Can use $1\{p_i \leq \alpha\}$ for a fixed i . But, better power if use all of them.
- Benferroni's test for global null:

$$\text{Reject } H_0 \text{ if } \min_i p_i \leq \frac{\alpha}{n}$$

- By union bound (no independence needed),

$$\mathbb{P}_{H_0}(\text{rejecting } H_0) = \mathbb{P}_{H_0}\left(\bigcup_{i=1}^n \{p_i \leq \alpha/n\}\right) \leq \sum_{i=1}^n \mathbb{P}_{H_0}(p_i \leq \alpha/n) = \alpha$$

- Exercise: Assuming p_i s are independent under H_0 , the exact the size of Benferroni's test is $1 - (1 - \frac{\alpha}{n})^n \rightarrow 1 - e^{-\alpha}$ as $n \rightarrow \infty$. Thus for large n and small alpha, size $\approx 1 - e^{-\alpha} \approx \alpha$, hence union bound is not bad in this case.

Fisher test for global null

- Fisher combination test:

$$\text{Reject } H_0 \text{ if } T_n := - \sum_{i=1}^n 2 \log p_i > \chi_{2n}^2(1 - \alpha)$$

Lemma 7

If $p_1, \dots, p_n$ are independent, then $T_n \sim \chi_{2n}^2$.

- Thus, assuming independence under H_0 , Fisher test has the correct size.

Simes test for global null

- Simes test:

$$\text{Reject } H_0 \text{ if } T_n := \min_i \left\{ p_{(i)}, \frac{n}{i} \right\} \leq \alpha$$

where $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(n)}$ is the order statistics of $p_1, \dots, p_n$.

Lemma 8

If $p_1, \dots, p_n$ are independent, then $T_n \sim U[0, 1]$.

- (Independence can be relaxed.)
- Thus, assuming independence under H_0 , Simes test has the correct size.
- Equivalent form of Simes test:

$$\text{Reject } H_0 \text{ if } p_{(i)} \leq \frac{i}{n} \alpha \text{ for some } i$$

- Less conservative than Benferroni's that reject if $p_{(1)} \leq \frac{1}{n} \alpha$.

Testing individual hypotheses

- In the gene expression example, we care about which genes are null/not null. We would like to test $H_{0,i} : \mu_i = 0$ versus $H_{1,i} : \mu_i = 1$ for all i .
- The problem has resemblance to classification.
- Interested in how we are doing in an [aggregate sense](#).
- We can think of having a decision problem

$$p \sim \mathbb{P}_\theta, \quad \text{where} \quad \theta \in \{0, 1\}^n. \quad (10)$$

- $\theta_i = 1$ iff $H_{0,i}$ is true.
- Global null corresponds to $\theta = 0$ (zero vector).
- Minimal assumption: When $\theta_i = 0$, we have $p_i \sim U[0, 1]$, i.e., the i th marginal of P_θ is uniform.

Terminology (shared with classification)

- Confusion matrix: Count how many combinations we have.
- For example if $\hat{\theta} \in \{0, 1\}^n$ is our guess for the hypotheses:

$$\text{TP} = \frac{1}{n} \sum_{i=1}^n 1\{\theta_i = 1, \hat{\theta}_i = 1\}$$

	positive (1) accepted	negative (0) rejected	total
true (1)	TP	TN	T
false (0)	FP	FN	F
	P	N	

- True here means $H_{0,i}$ is true.

- Alternative notation

	positive (1) accepted	negative (0) rejected	total
true (1)	U	V	n_0
false (0)	T	S	$n - n_0$
	$n - R$	R	n

- Familywise error rate (FWER):

$$\text{FWER}_\theta = \mathbb{P}_\theta(V \geq 1)$$

- A much less stringent criterion is the **False Discovery Rate (FDR)**.
- Consider the false discovery proportion (a random variable)

$$\text{FDP} = \frac{V}{\max(R, 1)} = \frac{V}{R} 1\{R > 0\}$$

- FDR is the expectation of FDP:

$$\text{FDR}_\theta = \mathbb{E}_\theta[\text{FDP}]$$

- Controlling FWER in a strong sense: control for all $\theta \in \{0, 1\}^n$.

Theorem 17

Benferroni's approach, where we test each hypothesis at level α/n , controls FWER at level α in a strong sense. In fact

$$\mathbb{E}_\theta[V] \leq \frac{n_0}{n} \alpha, \quad \forall \theta \in \{0, 1\}^n.$$

- $\mathbb{E}[V] = \sum_{i=1}^n \mathbb{P}(V \geq i)$ which holds for any nonnegative discrete variable.
- Hence, $\mathbb{E}[V] \geq \mathbb{P}(V \geq 1)$.
- Let $V_i = 1\{H_{i,0} \text{ is true but rejected}\} = 1\{\theta_i = 1, \hat{\theta}_i = 0\}$,

$$\mathbb{E}_\theta[V_i] = 1\{\theta_i = 1\} \mathbb{P}_\theta(\hat{\theta}_i = 0)$$

- Since $V = \sum_i V_i$,

$$\mathbb{E}_\theta[V] = \sum_i \mathbb{E}_\theta[V_i] = \sum_{i:\theta_i=1} \mathbb{P}_\theta(\hat{\theta}_i = 0) \leq \sum_{i:\theta_i=1} \frac{\alpha}{n} = \frac{n_0}{n} \alpha$$

(Here $\hat{\theta}_i$ is only based on p_i .)

- Benjamini-Hochberg (BH) procedure: Let i_0 be the largest i such that

$$p_{(i)} \leq \frac{i}{n}q$$

Reject all $H_{0,i}$ for $i \leq i_0$.

Theorem 18

Under independence of null hypotheses from each other and from the non-nulls, the BH procedure (uniformly) controls the FDR at level q . In fact,

$$FDR_{\theta}(\hat{\theta}_{BH}) = \frac{n_0}{n}q, \quad \forall \theta \in \{0, 1\}^n$$